



# Chapter 2 - Data, Measurements, and Data Preprocessing

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# 1. Data Types

- Data sets are made up of data objects.
- A **data object** represents an entity—in a sales database, the objects may be customers, store items, and sales; in a medical database, the objects may be patients; in a university database, the objects may be students, professors, and courses.
  - Data objects are typically described by attributes. Data objects can also be referred to as *samples*, *examples*, *instances*, *data points*, or *objects*.
  - If the data objects are stored in a database, they are *data tuples*. That is, the rows of a database correspond to the data objects, and the columns correspond to the attributes.
- An **attribute** is a data field, representing a characteristic or feature of a data object. The nouns *attribute*, *dimension*, *feature*, and *variable* are often used interchangeably in the literature.
- The **type** of an attribute is determined by the set of possible values—**nominal, binary, ordinal, or numeric**—the attribute can have.



# 1. Data Types

## Nominal Attributes

- Nominal means “relating to names.”
- The values of a **nominal attribute** are symbols or *names of things*.
- Each value represents some kind of category, code, or state, and so nominal attributes are also referred to as **categorical**. The values do not have any meaningful order. In computer science, the values are also known as enumerations.
- **Example 2.1. Nominal attributes.**
  - *hair\_color* attribute: possible values for *hair\_color* are *black*, *brown*, *blond*, *red*, *auburn*, *gray*, and *white*.
  - *marital\_status* attribute: can take on the values *single*, *married*, *divorced*, and *widowed*.
  - *occupation* attribute: with the values *teacher*, *dentist*, *programmer*, *farmer*, and so on.



# 1. Data Types

## Nominal Attributes

- Although we said that the values of a nominal attribute are symbols or “names of things,” it is possible to represent such symbols or “names” with numbers.
  - With *hair\_color*, for instance, we can assign a code of 0 for *black*, 1 for *brown*, and so on.
  - Another example is *customer\_ID*, with possible values that are all numeric.
- However, in such cases, the numbers are not intended to be used quantitatively.
  - That is, mathematical operations on values of nominal attributes are not meaningful. It makes no sense to subtract one customer ID number from another, unlike, say, subtracting an age value from another (where age is a numeric attribute).
  - Even though a nominal attribute may have integers as values, it is not considered a numeric attribute because the integers are not meant to be used quantitatively.
  - Because nominal attribute values do not have any meaningful order about them and are not quantitative, it makes no sense to find the mean (average) value or median (middle) value for such an attribute, given a set of objects. One thing that is of interest, however, is the attribute’s most commonly occurring value. This value, known as the *mode*, is one of the measures of central tendency.



# 1. Data Types

## Binary Attributes

- A **binary attribute** is a nominal attribute with only two categories or states: 0 or 1, where 0 typically means that the attribute is absent, and 1 means that it is present.
  - Binary attributes are referred to as **Boolean** if the two states correspond to *true* and *false*.
- A binary attribute is **symmetric** if both of its states are equally valuable and carry the same weight; that is, there is no preference on which outcome should be coded as 0 or 1.
  - One such example could be the attribute *gender* having the states *male* and *female*.
- A binary attribute is **asymmetric** if the outcomes of the states are not equally important, such as the *positive* and *negative* outcomes of a medical test for HIV. By convention, we code the most important outcome, which is usually the rarer one, by 1 (e.g., HIV positive) and the other by 0 (e.g., HIV negative).



# 1. Data Types

## Ordinal Attributes

- An **ordinal attribute** is an attribute with possible values that have a meaningful order or *ranking* among them, but the magnitude between successive values is not known.
- **Example 2.3. Ordinal attributes.** Suppose that *drink\_size* corresponds to the size of drinks available at a fast-food restaurant. This nominal attribute has three possible values: *small*, *medium*, and *large*.
  - The values have a meaningful sequence (which corresponds to increasing drink size); however, we cannot tell from the values *how much* bigger, say, a large is than a medium.
  - Other examples of ordinal attributes include *grade* (e.g., A+, A, A–, B+, and so on)
  - *professional\_rank*. Professional ranks can be enumerated in a sequential order: for example, *assistant*, *associate*, and *full* for professors.
- Ordinal attributes are useful for registering subjective assessments of qualities that cannot be measured objectively; thus ordinal attributes are often used in surveys for ratings.
  - In one survey, participants were asked to rate how satisfied they were as customers. Customer satisfaction had the following ordinal categories: 1: *very dissatisfied*, 2: *dissatisfied*, 3: *neutral*, 4: *satisfied*, and 5: *very satisfied*.



# 1. Data Types

## Ordinal Attributes

- Ordinal attributes may also be obtained from the discretization of numeric quantities by splitting the value range into a finite number of ordered categories as described in a later section on data reduction.
- The central tendency of an ordinal attribute can be represented by its mode and its median (the middle value in an ordered sequence), but the mean cannot be defined.
- Note that nominal, binary, and ordinal attributes are qualitative. That is, they describe a feature of an object without giving an actual size or quantity.
  - The values of such qualitative attributes are typically words representing categories.
  - If integers are used, they represent computer codes for the categories, as opposed to measurable quantities.





# 1. Data Types

## Numeric Attributes

- A **numeric attribute** is *quantitative*; that is, it is a measurable quantity, represented in integer or real values.
- Numeric attributes can be *interval-scaled* or *ratio-scaled*.
- **Interval-scaled Attributes**
  - **Interval-scaled attributes** are measured on a scale of equal-size units. The values of interval-scaled attributes have order and can be positive, 0, or negative. Thus, in addition to providing a ranking of values, such attributes allow us to compare and quantify the difference between values.
  - **Example 2.4. Interval-scaled attributes.** A *temperature* attribute is interval-scaled.
  - Because interval-scaled attributes are numeric, we can compute their mean value, in addition to the median and mode measures of central tendency.



# 1. Data Types

## Numeric Attributes

- **Ratio-scaled Attributes**

- A **ratio-scaled attribute** is a numeric attribute with an inherent zero-point. That is, if a measurement is ratio-scaled, we can speak of a value as being a multiple (or ratio) of another value.
- In addition, the values are ordered, and we can also compute the difference between values, as well as the mean, median, and mode.
- **Example 2.5. Ratio-scaled attributes.** Unlike temperatures in Celsius and Fahrenheit, the Kelvin (K) temperature scale has what is considered a true zero-point ( $0\text{ K} = -273.15\text{ }^{\circ}\text{C}$ ): It is the point at which all thermal motion ceases in the classical description of thermodynamics.
  - Other examples of ratio-scaled attributes include count attributes such as *years\_of\_experience* (e.g., the objects are employees) and *number\_of\_words* (e.g., the objects are documents). Additional examples include attributes to measure weight, height, and speed, and monetary quantities (e.g., you are 100 times richer with \$100 than with \$1).



# 1. Data Types

## Discrete vs. Continuous Attributes

- In our presentation, we have organized attributes into nominal, binary, ordinal, and numeric types. There are many ways to organize attribute types. The types are not mutually exclusive.
- A **discrete attribute** has a finite or countably infinite set of values, which may or may not be represented as integers.
- If an attribute is not discrete, it is **continuous**.
  - The terms numeric attribute and continuous attribute are often used interchangeably in the literature. (This can be confusing because, in the classic sense, continuous values are real numbers, whereas numeric values can be either integers or real numbers.) In practice, real values are represented using a finite number of digits. Continuous attributes are typically represented as floating-point variables.



## 2. Statistics of Data

- For data preprocessing to be successful, it is essential to have an overall picture of your data.
- Basic statistical descriptions can be used to identify properties of the data and highlight which data values should be treated as noise or outliers.



## 2. Statistics of Data

- Basic statistical descriptions
  - Measures of central tendency: measure the location of the middle or center of a data distribution
    - Intuitively speaking, given an attribute, where do most of its values fall?
    - The mean, median, mode, and midrange
  - Measures of dispersion of data: how are the data spread out?
    - Range, quartiles, interquartile range
    - Five-number summary, boxplots
    - Variance, standard deviation
  - Relations among multiple variables
    - Co-variance, correlation coefficient for numeric data
    - $\chi^2$  correlation test for nominal data
  - Graphic display of basic statistical description



## 2. Statistics of Data

### 2.1. Measuring the Central Tendency

- Suppose that we have some attribute  $X$ , like *salary*, which has been recorded for a set of objects. Let  $x_1, x_2, \dots, x_N$  be the set of  $N$  observed values or observations for  $X$ .
  - If we were to plot the observations for *salary*, **where would most of the values fall**? This gives us an idea of the central tendency of the data.
- Measures of central tendency include the mean, median, mode, and midrange

## 2. Statistics of Data

### 2.1. Measuring the Central Tendency

#### Mean

- **(Arithmetic) Mean**

- Numeric measure of the center of a set of data:

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + \dots + x_N}{N}$$

- **Example 2.6. Mean.** Suppose we have the following values for *salary* (in thousands of dollars), shown in ascending order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.

$$\bar{x} = \frac{30 + 36 + 47 + 50 + 52 + 52 + 56 + 60 + 63 + 70 + 70 + 110}{12} = 58$$

Thus, the mean salary is \$58,000.

## 2. Statistics of Data

### 2.1. Measuring the Central Tendency

#### Mean

- **Weighted arithmetic mean (weighted average)**
  - Sometimes, each value  $x_i$  in a set may be associated with a weight  $w_i$ . The weights reflect the significance, importance, or occurrence frequency attached to their respective values.

$$\bar{x} = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i} = \frac{w_1 x_1 + w_2 x_2 + \dots + w_N x_N}{w_1 + w_2 + \dots + w_N}$$



## 2. Statistics of Data

### 2.1. Measuring the Central Tendency Mean

- **Trimmed mean**

- Although the mean is the single most useful quantity for describing a data set, it is not always the best way of measuring the center of the data.
- A major problem with the mean is its sensitivity to extreme (e.g., outlier) values. Even a small number of extreme values can corrupt the mean.
- To offset the effect caused by a small number of extreme values, we can instead use the **trimmed mean**, which is the mean obtained after chopping off values at the high and low extremes.
  - For example, we can sort the values observed for salary and remove the top and bottom 2% before computing the mean.
  - We should avoid trimming too large a portion (such as 20%) at both ends, as this can result in the loss of valuable information.

## 2. Statistics of Data

### 2.1. Measuring the Central Tendency

#### Median

- For skewed (asymmetric) data, a better measure of the center of data is the **median**, which is the middle value in a set of ordered data values.
  - It is the value that separates the higher half of a data set from the lower half.
- In probability and statistics, the median generally applies to numeric data; however, we may extend the concept to ordinal data.
  - Suppose that a given data set of  $N$  values for an attribute  $X$  is sorted in ascending order.
    - If  $N$  is odd, then the median is the middle value of the ordered set.
    - If  $N$  is even, then the median is not unique; it is the two middlemost values and any value in between. If  $X$  is a numeric attribute in this case, by convention, the median is taken as the average of the two middlemost values.

## 2. Statistics of Data

### 2.1. Measuring the Central Tendency

#### Median

- **Example 2.7. Median.** Suppose we have the following values for salary (in thousands of dollars), shown in ascending order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110. The data are already sorted in ascending order.
  - There is an even number of observations (i.e., 12); therefore, the median is not unique.
  - It can be any value within the two middlemost values of 52 and 56 (that is, within the sixth and seventh values in the list). By convention, we assign the average of the two middlemost values as the median; that is,  $(52+56)/2 = 54$ . Thus, the median is \$54,000.
- The median is expensive to compute when we have a large number of observations.

## 2. Statistics of Data

### 2.1. Measuring the Central Tendency

#### Median

- For numeric attributes, however, we can easily *approximate* the value.
- Assume that data are grouped in intervals according to their  $x_i$  data values and that the frequency (i.e., number of data values) of each interval is known.
  - For example, employees may be grouped according to their annual salary in intervals such as \$10,001–20,000, \$20,001–50,000, and so on.
  - Let the interval that contains the median frequency be the *median interval*. We can approximate the median of the entire data set (e.g., the median salary) by interpolation using the formula:

$$median \approx L_1 + \left( \frac{N/2 - (\sum freq)_l}{freq_{median}} \right) \times width$$

- $L_1$  is the lower boundary of the median interval
- $N$  is the number of values in the entire data set
- $(\sum freq)_l$  is the sum of the frequencies of all of the intervals that are lower than the median interval
- $freq_{median}$  is the frequency of the median interval
- $width$  is the width of the median interval.

## 2. Statistics of Data

### 2.1. Measuring the Central Tendency

#### Mode

- *Mode* is another measure of central tendency.
- The **mode** for a set of data is the value that occurs most frequently compared to all neighboring values in the set.
  - Therefore, it can be determined for qualitative and quantitative attributes. It is possible for the greatest frequency to correspond to several different values, which results in more than one mode.
  - Data sets with one, two, or three modes are respectively called **unimodal**, **bimodal**, and **trimodal**. In general, a data set with two or more modes is **multimodal**.
- **Example 2.8. Mode.** Suppose we have the following values for salary (in thousands of dollars), shown in ascending order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.
  - The data are bimodal. The two modes are \$52,000 and \$70,000.

## 2. Statistics of Data

### 2.1. Measuring the Central Tendency

#### Mode

- For unimodal numeric data that are moderately skewed (asymmetrical), we have the following empirical relation:

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$$mean - mode \approx 3 \times (mean - median)$$

mean là số có thể + - \* /

median 0.

- This implies that the mode for unimodal frequency curves that are moderately skewed can easily be approximated if the mean and median values are known.

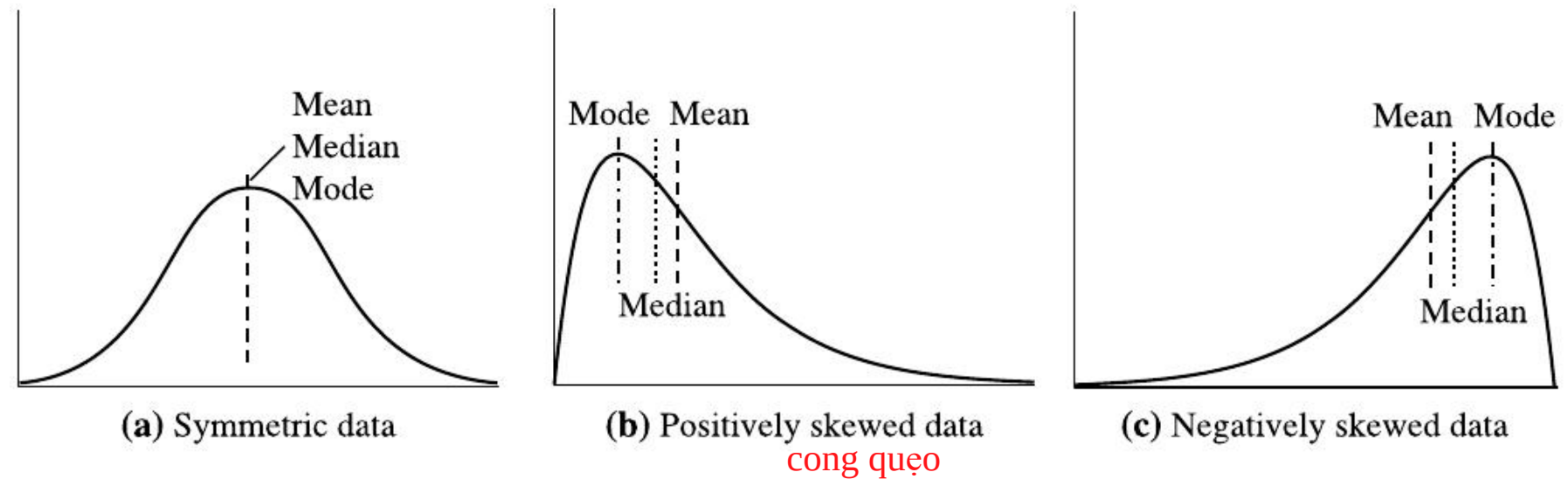
## 2. Statistics of Data

### 2.1. Measuring the Central Tendency

#### Mode

- In a unimodal frequency curve with perfect **symmetric** data distribution, the mean, median, and mode are all at the same center value, as shown in Fig. 2.1a.

- Data in most real applications are not symmetric.
- They may instead be either positively skewed, where the mode occurs at a value that is smaller than the median (Fig. 2.1b), or negatively skewed, where the mode occurs at a value greater than the median (Fig. 2.1c).



**FIGURE 2.1**

Mean, median, and mode of symmetric vs. positively and negatively skewed data.

## 2. Statistics of Data

### 2.1. Measuring the Central Tendency

#### Midrange

- The **midrange** can also be used to assess the central tendency of a numeric data set. It is the average of the largest and smallest values in the set.
- **Example 2.9. Midrange.** Suppose we have the following values for salary (in thousands of dollars), shown in ascending order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.
  - The midrange of the data is  $(30,000 + 110,000)/2 = \$70,000$ .





## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

sự gom tụ dữ liệu

- The measures include
  - Range, quantiles, quartiles, percentiles, and the interquartile range
  - The five-number summary, which can be displayed as a boxplot, is useful in identifying outliers.
  - Variance and standard deviation also indicate the spread of a data distribution.

## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

#### Range, Quantiles, Interquartile Range

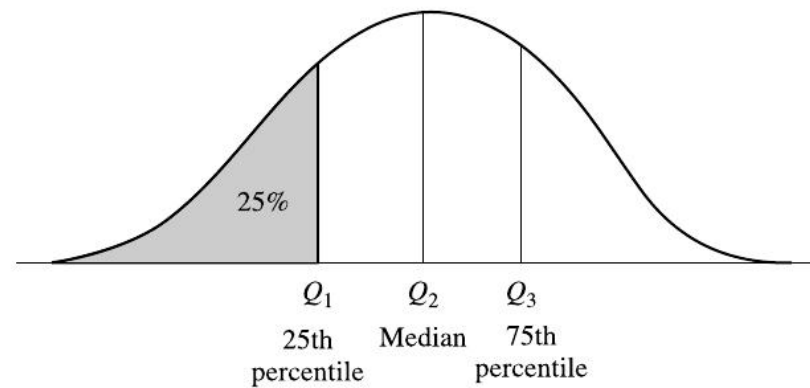
- Let  $x_1, x_2, \dots, x_N$  be a set of observations for some numeric attribute,  $X$ .
- The **range** of the set is the difference between the largest and smallest values.

## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

#### Range, Quantiles, Interquartile Range

- Suppose that the data for attribute  $X$  are sorted in ascending numeric order.
- Imagine that we can pick certain data points so as to split the data distribution into equal-size consecutive sets. These data points are called *quantiles*. **Quantiles** are points taken at regular intervals of a data distribution, dividing it into essentially equal-size consecutive sets.



**FIGURE 2.2**

A plot of the data distribution for some attribute  $X$ . The quantiles plotted are quartiles. The three quartiles divide the distribution into four equal-size consecutive subsets. The second quartile corresponds to the median.

## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

#### Range, Quantiles, Interquartile Range

- The  $k^{\text{th}}$   $q$ -quantile for a given data distribution is the value  $x$  such that at most  $k/q$  of the data values are less than  $x$  and at most  $(q - k)/q$  of the data values are more than  $x$ , where  $k$  is an integer such that  $0 < k < q$ . There are  $q - 1$   $q$ -quantiles.
- The 2-quantile is the data point dividing the lower and upper halves of the data distribution. It corresponds to the median.
- The 4-quantiles are the three data points that split the data distribution into four equal parts; each part represents one-fourth of the data distribution. They are more commonly referred to as **quartiles**.
- The 100-quantiles are more commonly referred to as **percentiles**; they divide the data distribution into 100 equal-size consecutive sets.
- The median, quartiles, and percentiles are the most widely used forms of quantiles.

## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

#### Range, Quantiles, Interquartile Range

- The quartiles give an indication of a distribution's center, spread, and shape.
  - The **first quartile**, denoted by  $Q_1$  , is the 25<sup>th</sup> percentile. It cuts off the lowest 25% of the data.
  - The **third quartile**, denoted by  $Q_3$  , is the 75<sup>th</sup> percentile—it cuts off the lowest 75% (or highest 25%) of the data.
  - The **second quartile** is the 50<sup>th</sup> percentile. As the median, it gives the center of the data distribution.

## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

#### Range, Quantiles, Interquartile Range

- The distance between the first and third quartiles is a simple measure of spread that gives the range covered by the middle half of the data. This distance is called the **interquartile range (IQR)** and is defined as

$$IQR = Q_3 - Q_1$$

- **Example 2.10. Interquartile range.** The quartiles are the three values that split the sorted data set into four equal parts. The data of Example 2.6 contain 12 observations, already sorted in ascending order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.
  - Since there are even number of elements on this list, the median of the list should be the mean of the center two elements, that is  $(\$52,000 + \$56,000)/2 = \$54,000$ .
  - Then the first quartile should be the mean of the 3rd and 4th elements, that is,  $(\$47,000 + \$50,000)/2 = \$48,500$
  - The 3rd quartile should be the mean of the 9th and 10th elements, that is,  $(\$63,000 + \$70,000)/2 = \$66,500$ .
  - Thus the interquartile range is  $IQR = \$66,500 - \$48,500 = \$18,000$ .

## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

#### Five-number Summary, Boxplots, and Outliers

- No single numeric measure of spread (e.g., IQR) is very useful for describing skewed distributions.
- In the symmetric distribution, the median (and other measures of central tendency) splits the data into equal-size halves.
- This does not occur for skewed distributions. Therefore it is more informative to also provide the two quartiles  $Q_1$  and  $Q_3$ , along with the median.
- A common rule of thumb for identifying suspected **outliers** is to single out values falling at least  $1.5 \times \text{IQR}$  above the third quartile or below the first quartile.

## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

#### Five-number Summary, Boxplots, and Outliers

- Because  $Q_1$ , the median, and  $Q_3$  together contain no information about the endpoints (e.g., tails) of the data, a fuller summary of the shape of a distribution can be obtained by providing the lowest and highest data values as well. This is known as the *five-number summary*.
- The **five-number summary** of a distribution consists of
  - The median ( $Q_2$ ),
  - The quartiles  $Q_1$  and  $Q_3$
  - The smallest and largest individual observations
- Written in the order of *Minimum*,  $Q_1$ , *Median*,  $Q_3$ , *Maximum*.



## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

#### Five-number Summary, Boxplots, and Outliers

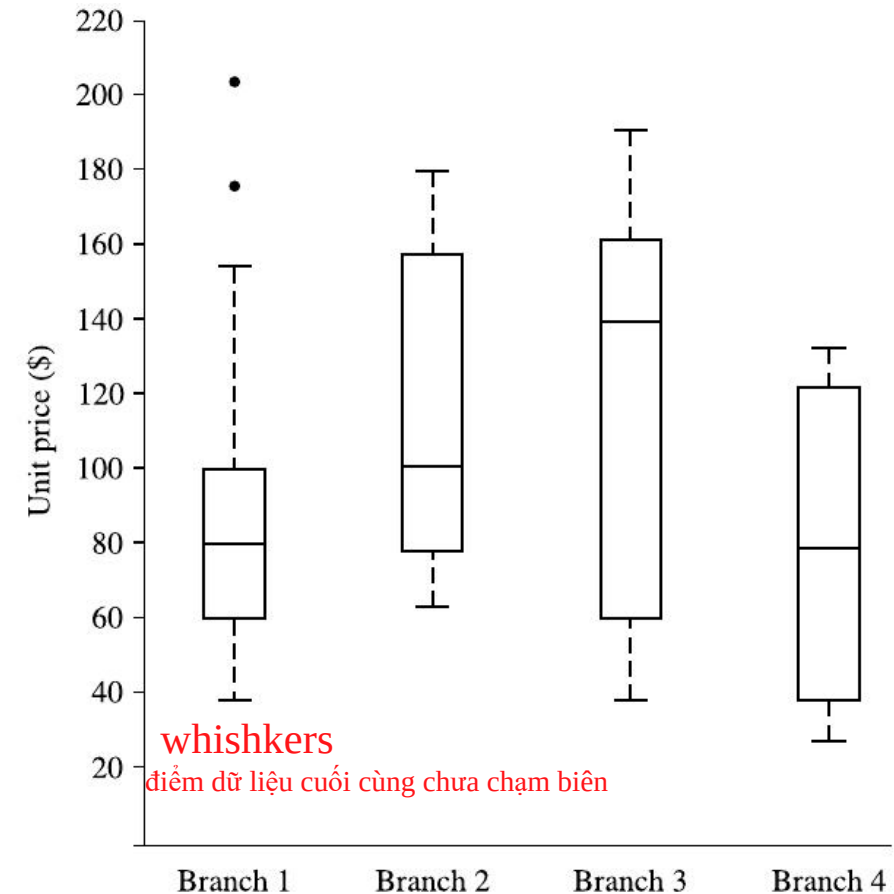
- **Boxplots** are a popular way of visualizing a distribution.
- A boxplot incorporates the five-number summary as follows:
  - Typically, the ends of the box are at the quartiles so that the box length is the interquartile range.
  - The median is marked by a line within the box.
  - Two lines (called *whiskers*) outside the box extend to the smallest (*Minimum*) and largest (*Maximum*) observations.
- When dealing with a moderate number of observations, it is worthwhile to plot potential outliers individually.
  - To do this in a boxplot, **the whiskers are extended to the extreme low and high observations only if these values are less than  $1.5 \times \text{IQR}$  beyond the quartiles. Otherwise, the whiskers terminate at the most extreme observations occurring within  $1.5 \times \text{IQR}$  of the quartiles.** The remaining cases are plotted individually. Boxplots can be used in the comparisons of several sets of compatible data.

## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

#### Five-number Summary, Boxplots, and Outliers

- **Example 2.11. Boxplot.** Fig. 2.3 shows boxplots for unit price data for items sold at four branches of an online store during a given time period.
  - For branch 1, we see that the median price of items sold is \$80, Q1 is \$60, and Q3 is \$100. Notice that two outlying observations for this branch were plotted individually, as their values of 175 and 202 are more than 1.5 times the IQR here of 40.



**FIGURE 2.3**

Boxplot for the unit price data for items sold at four branches of an online store during a given time period.

## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

#### Variance and Standard Deviation

- Variance and standard deviation are measures of data dispersion. They indicate how spread out a data distribution is.
  - A low standard deviation means that the data observations tend to be very close to the mean, whereas a high standard deviation indicates that the data are spread out over a large range of values.
- The **variance** of  $N$  observations,  $x_1, x_2, \dots, x_N$  (when  $N$  is large), for a numeric attribute  $X$  is

$$\overset{\text{sigma}}{\underset{\text{phương sai}}{\sigma^2}} = \frac{1}{N} \sum_{i=1}^N (\underset{\text{điểm cần tính}}{x_i} - \overset{\text{tâm}}{\bar{x}})^2 = \left( \frac{1}{N} \sum_{i=1}^N x_i^2 \right) - \bar{x}^2$$

trung bình của phương sai

khoảng cách của tập dữ liệu đến tâm là bao nhiêu  
lớn: cách xa

- The **standard deviation**,  $\sigma$ , of the observations is the square root of the variance,  $\sigma^2$ .

## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

#### Variance and Standard Deviation

- **Example 2.12. Variance and standard deviation.** Suppose we have the following values for salary (in thousands of dollars), shown in ascending order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.
  - In Example 2.6, we found  $\bar{x} = \$58,000$
  - To determine the variance and standard deviation of the data from that example, we set  $N = 12$  and

$$\sigma^2 = \frac{1}{12} (30^2 + 36^2 + 47^2 + \dots + 110^2) - 58^2 \approx 379.17$$

$$\sigma \approx \sqrt{379.17} \approx 19.47$$

## 2. Statistics of Data

### 2.2. Measuring the Dispersion of Data

#### Variance and Standard Deviation

- The basic properties of the standard deviation,  $\sigma$ , as a measure of spread are as follows:
  - $\sigma$  measures spread about the mean and should be considered only when the mean is chosen as the measure of center.
  - $\sigma = 0$  only when there is no spread, that is, when all observations have the same value. Otherwise,  $\sigma > 0$ .
- Importantly, an observation is unlikely to be more than several standard deviations away from the mean. Mathematically, using Chebyshev's inequality, it can be shown that at least  $\left(1 - \frac{1}{k^2}\right) \times 100\%$  of the observations are no more than  $k$  standard deviations from the mean. Therefore, the standard deviation is a good indicator of the spread of a data set.

- **Chebyshev's Inequality:**  $P(|X - \overset{\text{min}}{\mu}| \geq k\sigma) \leq \frac{1}{k^2}$ 
  - xác suất
  - sigma: độ lệch chuẩn
  - tổng = 1  
xác suất phải  $< 1/k^2$   
lấy cả bên âm và bên dương

## 2. Statistics of Data

### 2.3. Covariance and Correlation Analysis

#### Covariance of Numeric Data

hiệp phương sai

- In probability theory and statistics, correlation and covariance are two similar measures for assessing how much two attributes change together.
- Consider two numeric attributes  $A$  and  $B$  and a set of  $n$  real-valued observations  $\{(a_1, b_1), \dots, (a_n, b_n)\}$ .
- The mean values of  $A$  and  $B$ , respectively, are also known as the **expected values** on  $A$  and  $B$ , that is,

$$E(A) = \bar{A} = \frac{\sum_{i=1}^n a_i}{n}$$
$$E(B) = \bar{B} = \frac{\sum_{i=1}^n b_i}{n}$$

- The **covariance** between  $A$  and  $B$  is defined as

$$Cov(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}$$

- Mathematically, it can also be shown that

$$Cov(A, B) = E(A \cdot B) - \bar{A}\bar{B}$$

## 2. Statistics of Data

### 2.3. Covariance and Correlation Analysis

#### Covariance of Numeric Data

- For two attributes  $A$  and  $B$  that *tend to change together*, if a value  $a_i$  of  $A$  is larger than  $\bar{A}$ , then the corresponding value of  $b_i$  of attribute  $B$  is likely to be larger than  $\bar{B}$ . Therefore the covariance between  $A$  and  $B$  is *positive*.
  - On the other hand, if one of the attributes tends to be above its expected value when the other attribute is below its expected value, then the covariance of  $A$  and  $B$  is *negative*.
- If  $A$  and  $B$  are *independent* (i.e., they do not have correlation), then
  - $E(A \cdot B) = E(A) \cdot E(B)$ .
  - $\text{Cov}(A, B) = E(A \cdot B) - \bar{A}\bar{B} = E(A) \cdot E(B) - \bar{A}\bar{B} = 0$ .
  - However, the converse is not true. Some pairs of random variables (attributes) may have a covariance of 0 but are not independent. Only under some additional assumptions (e.g., the data follow multivariate normal distributions) does a covariance of 0 imply independence.

## 2. Statistics of Data

### 2.3. Covariance and Correlation Analysis

#### Covariance of Numeric Data

- **Example 2.13. Covariance analysis of numeric attributes.**

Consider Table 2.1, which presents a simplified example of stock prices observed at five time points for *AllElectronics* and *HighTech*, a high-tech company. If the stocks are affected by the same industry trends, will their prices rise or fall together?

**Table 2.1** Stock prices for *AllElectronics* and *HighTech*.

| Time point | AllElectronics | HighTech |
|------------|----------------|----------|
| t1         | 6              | 20       |
| t2         | 5              | 10       |
| t3         | 4              | 14       |
| t4         | 3              | 5        |
| t5         | 2              | 5        |



## 2. Statistics of Data

### 2.3. Covariance and Correlation Analysis

#### Covariance of Numeric Data

- $E(AllElectronics) = \frac{6+5+4+3+2}{5} = \$4$
- $E(HighTech) = \frac{20+10+14+5+5}{5} = \$10.80$
- $$Cov(AllElectotics, HighTech) = \frac{6 \times 20 + 5 \times 10 + 4 \times 14 + 3 \times 5 + 2 \times 5}{5} - 4 \times 10.80$$
$$= 50.2 - 43.2 = 7$$

**Table 2.1** Stock prices for *AllElectronics* and *HighTech*.

| Time point | AllElectronics | HighTech |
|------------|----------------|----------|
| t1         | 6              | 20       |
| t2         | 5              | 10       |
| t3         | 4              | 14       |
| t4         | 3              | 5        |
| t5         | 2              | 5        |

Therefore, given the positive covariance we can say that stock prices for both companies rise together.

## 2. Statistics of Data

### 2.3. Covariance and Correlation Analysis

#### Covariance of Numeric Data

- *Variance* is a special case of covariance, where the two attributes are identical (i.e., the covariance of an attribute with itself).

## 2. Statistics of Data

### 2.3. Covariance and Correlation Analysis

#### Correlation Coefficient for Numeric Data

- For numeric attributes, we can evaluate the correlation between two attributes,  $A$  and  $B$ , by computing the **correlation coefficient** (also known as **Pearson's product moment coefficient**, named after its inventor, Karl Pearson).

$$r_{A,B} = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n\sigma_A\sigma_B} = \frac{\sum_{i=1}^n (a_i b_i) - n\bar{A}\bar{B}}{n\sigma_A\sigma_B}$$

where  $n$  is the number of tuples,  $a_i$  and  $b_i$  are the respective values of  $A$  and  $B$  in tuple  $i$ .

- Note that  $-1 \leq r_{A,B} \leq +1$ .
  - If  $r_{A,B} > 0$ , then  $A$  and  $B$  are *positively correlated*, meaning that the values of  $A$  increase as the values of  $B$  increase. The higher the value, the stronger the correlation (i.e., the more each attribute implies the other). Hence, a higher value may indicate that  $A$  (or  $B$ ) may be removed as a redundancy.
  - If the resulting value is equal to 0, then  $A$  and  $B$  are independent, and there is no correlation between them.
  - If the resulting value is less than 0, then  $A$  and  $B$  are *negatively correlated*, where the values of one attribute increase as the values of the other attribute decrease.
  - Note that correlation does not imply causality. That is, if  $A$  and  $B$  are correlated, this does not necessarily imply that  $A$  causes  $B$  or that  $B$  causes  $A$ .

## 2. Statistics of Data

### 2.3. Covariance and Correlation Analysis

#### $\chi^2$ Correlation Test for Nominal Data

- For nominal data, a correlation relationship between two attributes,  $A$  and  $B$ , can be discovered by a  $\chi^2$  (**chi-square**) test.
- Suppose  $A$  has  $c$  distinct values, namely,  $a_1, a_2, \dots, a_c$ , and  $B$  has  $r$  distinct values, namely,  $b_1, b_2, \dots, b_r$ . The data tuples described by  $A$  and  $B$  can be shown as a **contingency table**, with the  $c$  values of  $A$  making up the columns and the  $r$  values of  $B$  making up the rows.
- Let  $(A_i, B_j)$  denote the joint event that attribute  $A$  takes on value  $a_i$  and attribute  $B$  takes on value  $b_j$ , that is, where  $(A = a_i, B = b_j)$ . Each and every possible  $(A_i, B_j)$  joint event has its own cell (or slot) in the table.
- The  $\chi^2$  value (also known as the Pearson  $\chi^2$  statistic) is computed as

$$\chi^2 = \sum_{i=1}^c \sum_{j=1}^r \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

where  $o_{ij}$  is the *observed frequency* (i.e., actual count) of the joint event  $(A_i, B_j)$  and  $e_{ij}$  is the *expected frequency* of  $(A_i, B_j)$ , which can be computed as

$$e_{ij} = \frac{\text{count}(A = a_i) \times \text{count}(B = b_j)}{n}$$

where  $n$  is the number of data tuples,  $\text{count}(A = a_i)$  is the number of tuples having value  $a_i$  for  $A$ , and  $\text{count}(B = b_j)$  is the number of tuples having value  $b_j$  for  $B$ .

## 2. Statistics of Data

### 2.3. Covariance and Correlation Analysis

#### $\chi^2$ Correlation Test for Nominal Data

- The  $\chi^2$  statistic tests the hypothesis that  $A$  and  $B$  are independent, that is, there is no correlation between them.
- The test is based on a significance level, with  $(r - 1) \times (c - 1)$  degrees of freedom.
- If the hypothesis can be rejected, then we say that  $A$  and  $B$  are statistically correlated.

## 2. Statistics of Data

### 2.3. Covariance and Correlation Analysis

#### $\chi^2$ Correlation Test for Nominal Data

- **Example 2.14. Correlation analysis of nominal attributes using  $\chi^2$ .** Suppose that a group of 1500 people was surveyed. The gender of each person was noted. Each person was polled as to whether his or her preferred type of reading material was fiction or nonfiction. Thus, we have two attributes, *gender* and *preferred\_reading*.
- The observed frequency (or count) of each possible joint event is summarized in the contingency table shown in Table 2.2, where the numbers in parentheses are the expected frequencies.

**Table 2.2** Example 2.1's  $2 \times 2$  contingency table data.

|                    | Male     | Female     | Total |
|--------------------|----------|------------|-------|
| <i>fiction</i>     | 250 (90) | 200 (360)  | 450   |
| <i>non_fiction</i> | 50 (210) | 1000 (840) | 1050  |
| Total              | 300      | 1200       | 1500  |

*Note: Are gender and preferred\_reading correlated?*

## 2. Statistics of Data

### 2.3. Covariance and Correlation Analysis

#### $\chi^2$ Correlation Test for Nominal Data

- $$e_{11} = \frac{\text{count}(\text{male}) \times \text{count}(\text{fiction})}{n} = \frac{300 \times 450}{1500} = 90$$

- ...

- $$\chi^2 = \frac{(250-90)^2}{90} + \frac{(50-210)^2}{210} + \frac{(200-260)^2}{360} + \frac{(1000-840)^2}{840} = 507.93$$

- For this  $2 \times 2$  table, the degrees of freedom are  $(2 - 1) \times (2 - 1) = 1$ . For 1 degree of freedom, the  $\chi^2$  value needed to reject the hypothesis at the 0.001 significance level is 10.828.

|    | 0.10  | 0.05   | 0.02   | 0.01   | 0.001  |
|----|-------|--------|--------|--------|--------|
| df |       |        |        |        |        |
| 1  | 2.706 | 3.841  | 5.412  | 6.635  | 10.828 |
| 2  | 4.605 | 5.991  | 7.824  | 9.210  | 13.816 |
| 3  | 6.251 | 7.815  | 9.837  | 11.345 | 16.266 |
| 4  | 7.779 | 9.488  | 11.668 | 13.277 | 18.467 |
| 5  | 9.236 | 11.070 | 13.280 | 15.086 | 20.515 |

**Table 2.2** Example 2.1's  $2 \times 2$  contingency table data.

|                    | Male     | Female     | Total |
|--------------------|----------|------------|-------|
| <i>fiction</i>     | 250 (90) | 200 (360)  | 450   |
| <i>non_fiction</i> | 50 (210) | 1000 (840) | 1050  |
| Total              | 300      | 1200       | 1500  |

Note: Are gender and preferred\_reading correlated?

- Since our computed value is above this, we can reject the hypothesis that *gender* and *preferred\_reading* are independent and conclude that the two attributes are (strongly) correlated for the given group of people.



## 2. Statistics of Data

### 2.4. Graphic Displays of Basic Statistics of Data

- Graphic displays of basic statistical descriptions
  - Univariate distributions (data for one attribute)
    - Quantile plots, quantile-quantile plots, histograms
  - Bivariate distributions (involving two attributes)
    - Scatter plots



## 2. Statistics of Data

### 2.4. Graphic Displays of Basic Statistics of Data

#### Quantile Plot

- A **quantile plot** is a simple and effective way to have a first look at a univariate data distribution.
  - First, it displays all of the data for the given attribute (allowing a user to assess both the overall behavior and unusual occurrences).
  - Second, it plots quantile information.
- Let  $x_i, i = 1 \dots N$ , be the data sorted in ascending order so that  $x_1$  is the smallest observation and  $x_N$  is the largest for some ordinal or numeric attribute  $X$ .
- Each observation,  $x_i$ , is paired with a percentage,  $f_i$ , which indicates that approximately  $f_i \times 100\%$  of the data are below the value,  $x_i$ . Note that the 0.25 quantile corresponds to quartile  $Q_1$ , the 0.50 quantile is the median, and the 0.75 quantile is  $Q_3$ .

- Let

$$f_i = \frac{i - 0.5}{N}$$

số phần tử

- These numbers increase in equal steps of  $1/N$ , ranging from  $\frac{1}{2N}$  (which is slightly above 0) to  $1 - \frac{1}{2N}$  (which is slightly below 1). On a quantile plot,  $x_i$  is graphed against  $f_i$ .

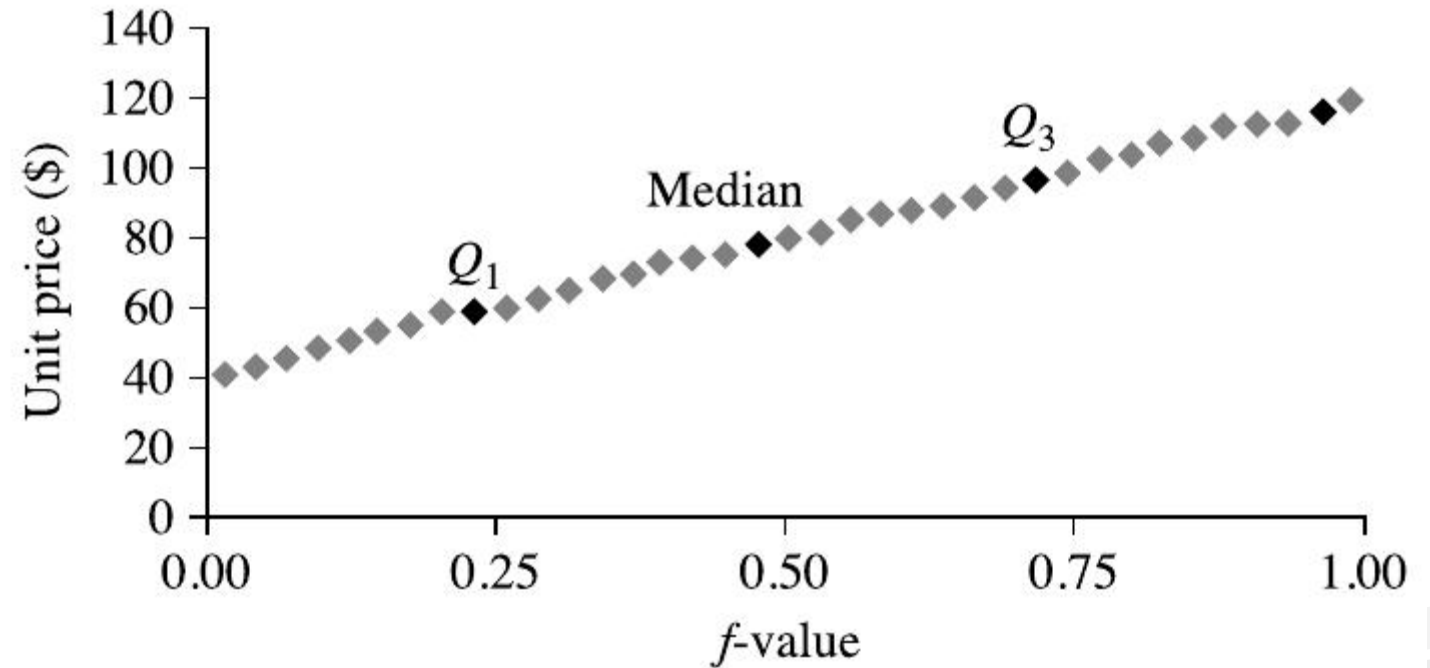
## 2. Statistics of Data

### 2.4. Graphic Displays of Basic Statistics of Data

#### Quantile Plot

**Table 2.3** A set of unit price data for items sold at a branch of the online store.

| Unit price (\$) | Count of items sold |
|-----------------|---------------------|
| 40              | 275                 |
| 43              | 300                 |
| 47              | 250                 |
| ⋮               | ⋮                   |
| 74              | 360                 |
| 75              | 515                 |
| 78              | 540                 |
| ⋮               | ⋮                   |
| 115             | 320                 |
| 117             | 270                 |
| 120             | 350                 |



## 2. Statistics of Data

### 2.4. Graphic Displays of Basic Statistics of Data

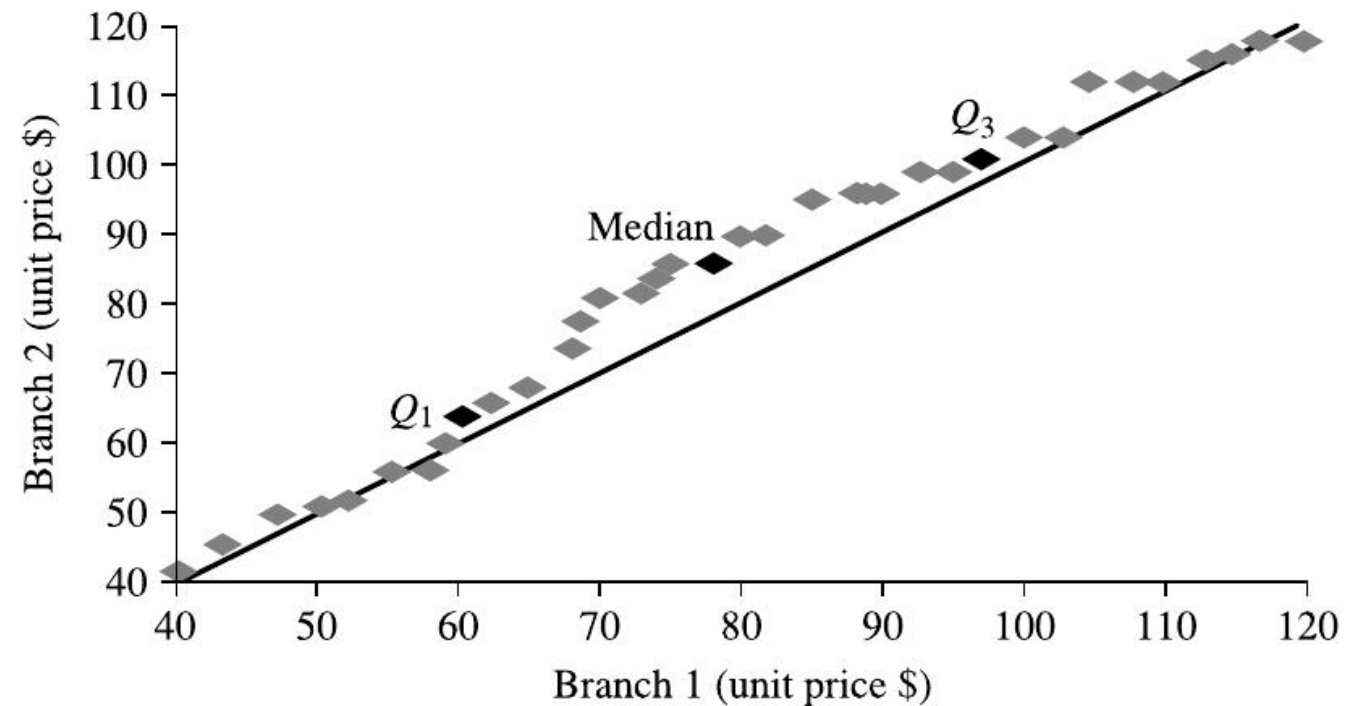
#### Quantile-Quantile Plot

- A **quantile-quantile plot**, or **q-q plot**, graphs the quantiles of one univariate distribution against the corresponding quantiles of another.
- It is a powerful visualization tool in that it allows the user to view whether there is a shift in going from one distribution to another.
- Suppose that we have two sets of observations for the attribute or variable *unit price*, taken from two different branch locations.
- Let  $x_1, \dots, x_N$  be the data from the first branch, and  $y_1, \dots, y_M$  be the data from the second, where each data set is sorted in ascending order.
- If  $M = N$  (i.e., the number of points in each set is the same), then we simply plot  $y_i$  against  $x_i$ , where  $y_i$  and  $x_i$  are both  $(i - 0.5)/N$  quantiles of their respective data sets.
- If  $M < N$  (i.e., the second branch has fewer observations than the first), there can be only  $M$  points on the q-q plot. Here,  $y_i$  is the  $(i - 0.5)/M$  quantile of the  $y$  data, which is plotted against the  $(i - 0.5)/M$  quantile of the  $x$  data. This computation typically involves interpolation.

## 2. Statistics of Data

### 2.4. Graphic Displays of Basic Statistics of Data

#### Quantile-Quantile Plot



**FIGURE 2.5**

A q-q plot for unit price data from two branches of the online store.

## 2. Statistics of Data

### 2.4. Graphic Displays of Basic Statistics of Data

#### Histograms (Frequency Histograms)

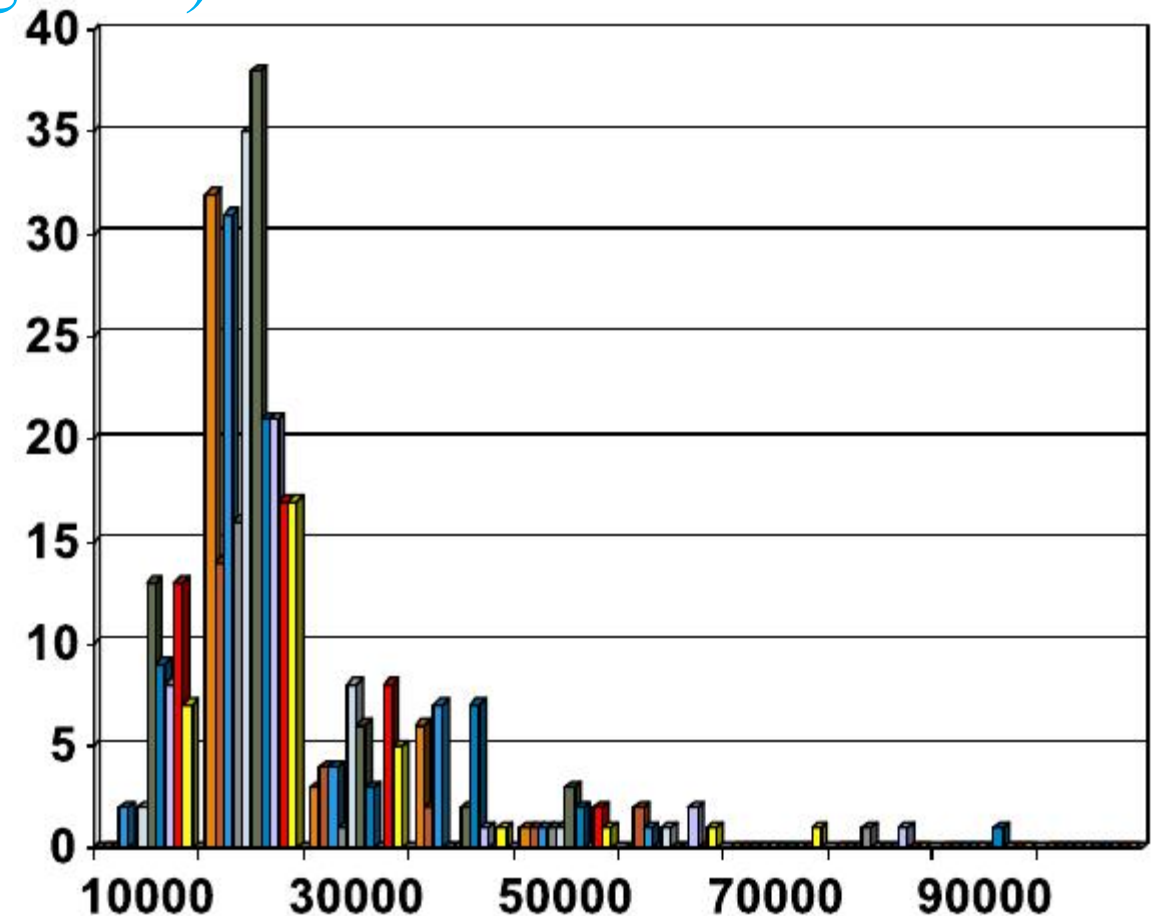
- Plotting histograms is a graphical method for summarizing the distribution of a given attribute,  $X$ .
- According to the number of poles desired in the chart, the range of values for  $X$  is partitioned into a set of disjoint consecutive subranges.
- The subranges, referred to as *buckets* or *bins*, are disjoint subsets of the data distribution for  $X$ . The range of a bucket is known as the **width**. Typically, the buckets are of equal width.
  - For example, a price attribute with a value range of \$1–\$200 (rounded up to the nearest dollar) can be partitioned into subranges 1–20, 21–40, 41–60, and so on. For each subrange, a bar is drawn with a height that represents the total count of items observed within the subrange.

## 2. Statistics of Data

### 2.4. Graphic Displays of Basic Statistics of Data

#### Histograms (Frequency Histograms)

- **Example 2.17. Histogram.** Fig. 2.6 shows a histogram for a data set on research award distribution for a region, where buckets (or bins) are defined by equal-width ranges representing \$1000 increments, and the frequency is the number of research awards in the corresponding buckets.



**FIGURE 2.6**

A histogram on research award distribution for a region.

## 2. Statistics of Data

### 2.4. Graphic Displays of Basic Statistics of Data

#### Scatter Plots and Data Correlation

- A **scatter plot** is one of the most effective graphical methods for determining whether there appears *to be a relationship, pattern, or trend* between two numeric attributes.
- To construct a scatter plot, each pair of values is treated as a pair of coordinates in an algebraic sense and plotted as points in the plane.

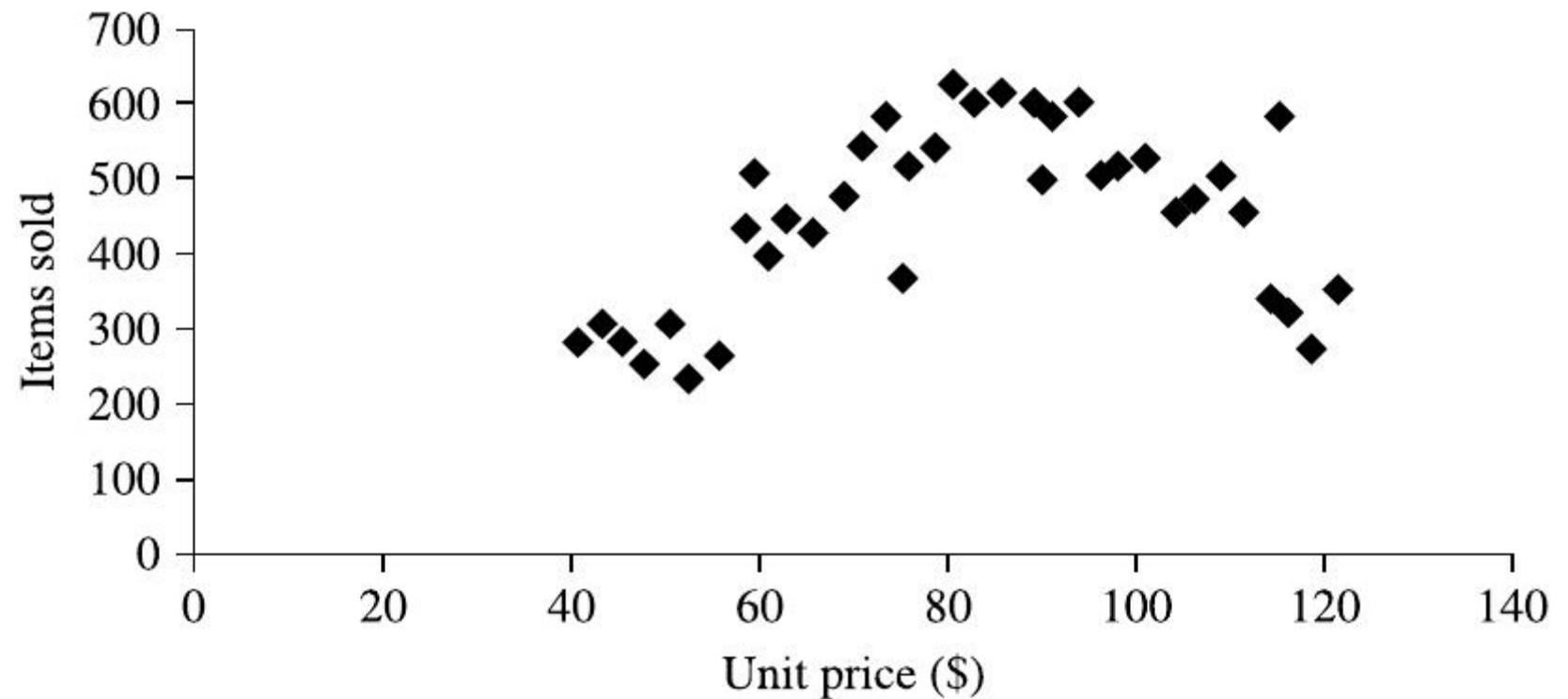
## 2. Statistics of Data

### 2.4. Graphic Displays of Basic Statistics of Data

#### Scatter Plots and Data Correlation

**Table 2.3** A set of unit price data for items sold at a branch of the online store.

| Unit price (\$) | Count of items sold |
|-----------------|---------------------|
| 40              | 275                 |
| 43              | 300                 |
| 47              | 250                 |
| ⋮               | ⋮                   |
| 74              | 360                 |
| 75              | 515                 |
| 78              | 540                 |
| ⋮               | ⋮                   |
| 115             | 320                 |
| 117             | 270                 |
| 120             | 350                 |





## 2. Statistics of Data

### 2.4. Graphic Displays of Basic Statistics of Data

#### Scatter Plots and Data Correlation

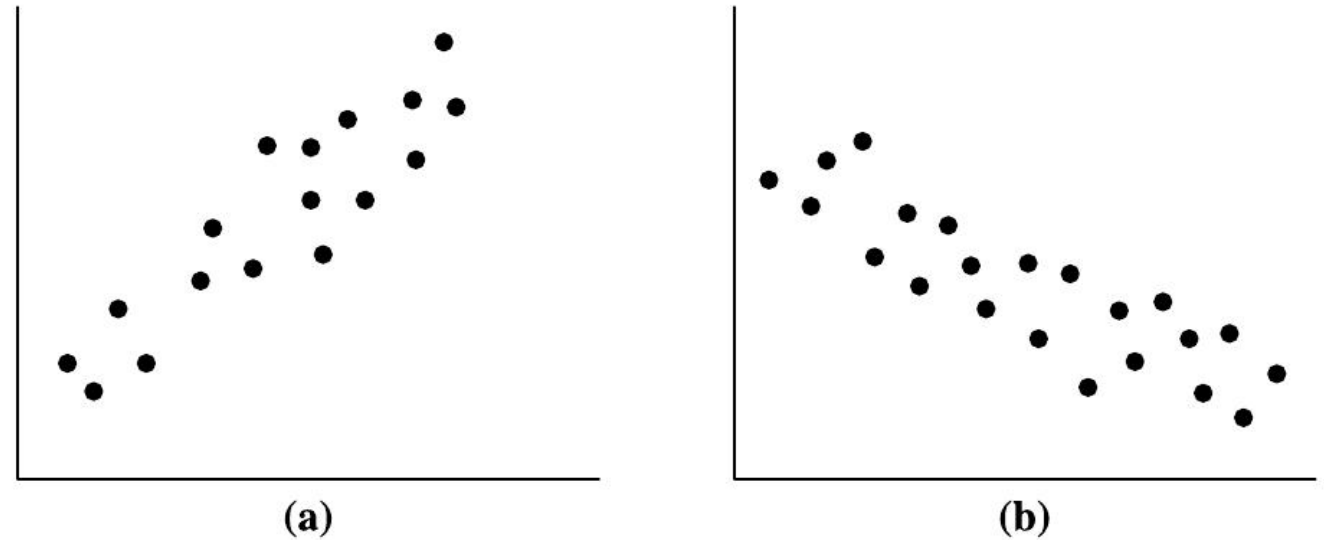
- The scatter plot is a useful method for providing a first look at bivariate data to see clusters of points and outliers, or to explore the possibility of correlation relationships.
- Two attributes,  $X$  and  $Y$ , are **correlated** if the knowledge of one attribute enables to predict the other with some accuracy.
  - Correlations can be positive, negative, or null (uncorrelated).

## 2. Statistics of Data

### 2.4. Graphic Displays of Basic Statistics of Data

#### Scatter Plots and Data Correlation

- If the plotted points pattern slopes from lower left to upper right, this means that the values of  $X$  increase as the values of  $Y$  increase, suggesting a *positive correlation* (Fig. 2.8a).
- If the pattern of plotted points slopes from upper left to lower right, the values of  $X$  increase as the values of  $Y$  decrease, suggesting a *negative correlation*.



**FIGURE 2.8**

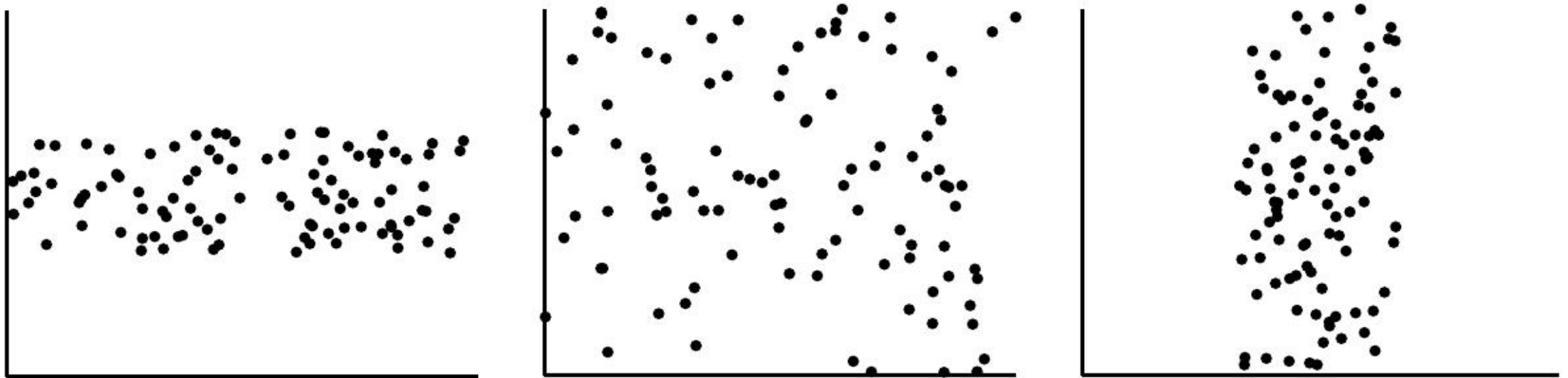
Scatter plots can be used to find (a) positive or (b) negative correlations between attributes.

## 2. Statistics of Data

### 2.4. Graphic Displays of Basic Statistics of Data

#### Scatter Plots and Data Correlation

0 có quan hệ gì heetttt



**FIGURE 2.9**

Three cases where there is no observed correlation between the two plotted attributes in each of the data sets.



### 3. Similarity and Distance Measurements

- This section presents similarity and dissimilarity measures, which are referred to as measures of *proximity*.
- Similarity and dissimilarity are related.
- A similarity measure for two objects,  $i$  and  $j$ , will typically return value 0 if the objects are completely *unlike*. The higher the similarity value, the greater the similarity between objects. (Typically, a value of 1 indicates complete similarity, that is, the objects are identical.)
- A dissimilarity measure works the opposite way. It returns a value of 0 if the objects are the same (and therefore, far from being dissimilar). The higher the dissimilarity value, the more dissimilar the two objects are.



## 3. Similarity and Distance Measurements

### 3.1 Data Matrix vs. Dissimilarity Matrix

- Suppose that we have  $n$  objects (e.g., persons, items, or courses) described by  $p$  attributes (also called *measurements* or *features*, such as age, height, weight, or gender).
- The objects are  $x_1 = (x_{11}, x_{12}, \dots, x_{1p})$ ,  $x_2 = (x_{21}, x_{22}, \dots, x_{2p})$ , and so on, where  $x_{ij}$  is the value for object  $x_i$  of the  $j$ th attribute.
  - For brevity, we hereafter refer to object  $x_i$  as object  $i$ . The objects may be tuples in a relational database and are also referred to as *data samples* or *feature vectors*.

### 3. Similarity and Distance Measurements

#### 3.1 Data Matrix vs. Dissimilarity Matrix

- **Data Matrix** (or *object-by-attribute structure*): This structure stores the  $n$  data objects in the form of a relational table or an  $n$ -by- $p$  matrix ( $n$  objects  $\times$   $p$  attributes)

$$\begin{bmatrix} x_{11} & \dots & x_{1p} \\ \dots & \dots & \dots \\ x_{n1} & \dots & x_{np} \end{bmatrix}$$

- **Dissimilar Matrix** (or *object-by-object structure*): This structure stores a collection of proximities that are available for all pairs of  $n$  objects. It is often represented by an  $n$ -by- $n$  table:

đo khoảng cách giữa các phần tử với nhau

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ d(n,1) & d(n,2) & \dots & \dots & 0 \end{bmatrix}$$

where  $d(i, j)$  is the measured **dissimilarity** or “difference” between objects  $i$  and  $j$ . In general,  $d(i, j)$  is a nonnegative number that is close to 0 when objects  $i$  and  $j$  are highly similar or “near” each other, and becomes larger the more they differ.

Note that  $d(i, i) = 0$ ; that is, the difference between an object and itself is 0. Furthermore,  $d(i, j) = d(j, i)$ .



## 3. Similarity and Distance Measurements

### 3.1 Data Matrix vs. Dissimilarity Matrix

- Measures of similarity can often be expressed as a function of measures of dissimilarity.

$$\underset{\text{similarity}}{sim(i, j)} = 1 - \underset{\text{dissimilarity}}{d(i, j)}$$

where  $sim(i, j)$  is the similarity between objects  $i$  and  $j$ .



## 3. Similarity and Distance Measurements

### 3.2. Proximity Measures for Nominal Attributes

- A nominal attribute can take on two or more states. For example, *map\_color* is a nominal attribute that may have, say, five states: *red*, *yellow*, *green*, *pink*, and *blue*.
- Let the number of states of a nominal attribute be  $M$ . The states can be denoted by letters, symbols, or a set of integers, such as  $1, 2, \dots, M$ . Notice that such integers are used just for data handling and do not represent any specific ordering.
- “*How is dissimilarity computed between objects described by nominal attributes?*” The dissimilarity between two objects  $i$  and  $j$  can be computed based on the ratio of mismatches:

$$d(i, j) = \frac{p - m}{p}$$

số thuộc tính giống nhau  
số cột, số attribute, thuộc tính

where  $m$  is the number of matches (i.e., the number of attributes for which  $i$  and  $j$  are in the same state), and  $p$  is the total number of attributes describing the objects. Weights can be assigned to increase the effect of  $m$  or to assign greater weight to the matches in attributes having a larger number of states.



# 3. Similarity and Distance Measurements

## 3.2. Proximity Measures for Nominal Attributes

- Example 2.18. Dissimilarity between nominal attributes.** Suppose that we have the sample data of Table 2.4, except that only the *object-identifier* and the attribute *test-1* are available, where *test-1* is nominal. Let's compute the dissimilarity matrix:

| Object Identifier | Test-1 (nominal) |
|-------------------|------------------|
| 1                 | code A           |
| 2                 | code B           |
| 3                 | code C           |
| 4                 | code A           |

vì có 1 test nên  $p = 1$

$$\begin{bmatrix} 0 & & & \\ d(2,1) & 0 & & \\ d(3,1) & d(3,2) & 0 & \\ d(4,1) & d(4,2) & d(4,3) & 0 \end{bmatrix} \text{ dissimilarity}$$

$$\begin{bmatrix} 0 & & & \\ 1-0/1=1 & 1 & 0 & \\ 1-0/1=1 & 1 & 1 & 0 \\ 1-1/1=0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Since here we have one nominal attribute, *test-1*, we set  $p = 1$  so that  $d(i, j)$  evaluates to 0 if objects  $i$  and  $j$  match, and 1 if the objects differ.

- Alternatively, similarity can be computed as  $sim(i, j) = 1 - d(i, j) = m / p$

## 3. Similarity and Distance Measurements

### 3.3. Proximity Measures for Binary Attributes

- Recall that a binary attribute has only one of two states, 0 and 1, where 0 means that the attribute is absent, and 1 means that it is present.
- “So, how can we compute the dissimilarity between two binary attributes?”*
- One approach involves computing a dissimilarity matrix from the given binary data. If all binary attributes are thought of as having the same weight, we have the  $2 \times 2$  contingency table, where  $q$  is the number of attributes that equal 1 for both objects  $i$  and  $j$ ,  $r$  is the number of attributes that equal 1 for object  $i$  but equal 0 for object  $j$ ,  $s$  is the number of attributes that equal 0 for object  $i$  but equal 1 for object  $j$ , and  $t$  is the number of attributes that equal 0 for both objects  $i$  and  $j$ . The total number of attributes is  $p$ , where  $p = q + r + s + t$ .

xét 2 object có  $q$  thuộc tính cả  $i, j$  đều là 1  
 $s$  thuộc tính mà  $i = 0, j = 1$   
 $r$  thuộc tính mà  $i = 1, j = 0$   
 $t$  thuộc tính mà cả 2 đều = 0  
 $p$  là số lượng thuộc tính

**Table 2.5 Contingency table for binary attributes.**

|            |   | Object $j$ |         |         |
|------------|---|------------|---------|---------|
|            |   | 1          | 0       | sum     |
| Object $i$ | 1 | $q$        | $r$     | $q + r$ |
|            | 0 | $s$        | $t$     | $s + t$ |
| sum        |   | $q + s$    | $r + t$ | $p$     |



## 3. Similarity and Distance Measurements

### 3.3. Proximity Measures for Binary Attributes

- Recall that for symmetric binary attributes, each state is equally valuable. Dissimilarity that is based on symmetric binary attributes is called symmetric binary dissimilarity.
- If objects  $i$  and  $j$  are described by symmetric binary attributes, then the dissimilarity between  $i$  and  $j$  is

$$d(i, j) = \frac{r + s}{q + r + s + t}$$



## 3. Similarity and Distance Measurements

### 3.3. Proximity Measures for Binary Attributes

- For asymmetric binary attributes, the two states are not equally important, such as the *positive* (1) and *negative* (0) outcomes of a disease test.
- Given two asymmetric binary attributes, the agreement of two 1s (a positive match) is then considered more significant than that of two 0s (a negative match). Therefore such binary attributes are often considered “monary” (having one state).
- The dissimilarity based on these attributes is called **asymmetric binary dissimilarity**, where the number of negative matches,  $t$ , is considered unimportant and is thus ignored in the following computation:

$$d(i, j) = \frac{r + s}{q + r + s}$$

- The asymmetric binary similarity between the objects  $i$  and  $j$  can be computed as

$$sim(i, j) = \frac{q}{q + r + s} = 1 - d(i, j)$$

- The coefficient  $sim(i, j)$  above is called the **Jaccard coefficient**.
- When both symmetric and asymmetric binary attributes occur in the same data set, the approach for mixed attributes described later.

## 3. Similarity and Distance Measurements

### 3.3. Proximity Measures for Binary Attributes

- **Example 2.19. Dissimilarity between binary attributes.** Suppose that a patient record table (Table 2.6) contains the attributes *name*, *gender*, *fever*, *cough*, *test-1*, *test-2*, *test-3*, and *test-4*, where *name* is an object identifier, *gender* is a symmetric binary attribute, and the remaining attributes are asymmetric binary.
- For asymmetric binary attribute values, let the values *Y* (*yes*) and *P* (*positive*) be set to 1, and the value *N* (*no* or *negative*) be set to 0. Suppose that the distance between objects (patients) is computed based only on the asymmetric binary attributes.

**Table 2.6 Relational table where patients are described by binary attributes.**

| Name | Gender | Fever | Cough | Test-1 | Test-2 | Test-3 | Test-4 |
|------|--------|-------|-------|--------|--------|--------|--------|
| Jack | M      | Y     | N     | P      | N      | N      | N      |
| Jim  | M      | Y     | Y     | N      | N      | N      | N      |
| Mary | F      | Y     | N     | P      | N      | P      | N      |
| ⋮    | ⋮      | ⋮     | ⋮     | ⋮      | ⋮      | ⋮      | ⋮      |

### 3. Similarity and Distance Measurements

#### 3.3. Proximity Measures for Binary Attributes

- Suppose that the distance between objects (patients) is computed based only on the *asymmetric binary attributes*.

**Table 2.6** Relational table where patients are described by binary attributes.

| Name | Gender | Fever | Cough | Test-1 | Test-2 | Test-3 | Test-4 |
|------|--------|-------|-------|--------|--------|--------|--------|
| Jack | M      | Y     | N     | P      | N      | N      | N      |
| Jim  | M      | Y     | Y     | N      | N      | N      | N      |
| Mary | F      | Y     | N     | P      | N      | P      | N      |
| ⋮    | ⋮      | ⋮     | ⋮     | ⋮      | ⋮      | ⋮      | ⋮      |

$$d(Jack, Jim) = \frac{1 + 1}{1 + 1 + 1} = 0.67,$$

$$d(Jack, Mary) = \frac{0 + 1}{2 + 0 + 1} = 0.33,$$

$$d(Jim, Mary) = \frac{1 + 2}{1 + 1 + 2} = 0.75.$$



### 3. Similarity and Distance Measurements

#### 3.4. Dissimilarity of Numeric Data: Minkowski Distance

- Distance measures: *Euclidean, Manhattan, Minkowski distances*
- In some cases, the data are normalized before applying distance calculations. This involves transforming the data to fall within a smaller or common range, such as  $[-1.0, 1.0]$  or  $[0.0, 1.0]$ .
  - Normalizing the data attempts to give all attributes an equal weight.
- The most popular distance measure is **Euclidean distance** (i.e., straight line or “as the crow flies”). Let  $i = (x_{i1}, x_{i2}, \dots, x_{ip})$  and  $j = (x_{j1}, x_{j2}, \dots, x_{jp})$  be two objects described by  $p$  numeric attributes. The Euclidean distance between objects  $i$  and  $j$  is defined as

$$d(i, j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{ip} - x_{jp})^2}$$



## 3. Similarity and Distance Measurements

### 3.4. Dissimilarity of Numeric Data: Minkowski Distance

- Another well-known measure is the **Manhattan** (or **city block**) **distance**, named so because it is the distance in blocks between any two points in a city:

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ip} - x_{jp}|$$

- Both the Euclidean and the Manhattan distance satisfy the following mathematical properties:
  - **Nonnegativity**:  $d(i, j) \geq 0$ : Distance is a nonnegative number.
  - **Identity of indiscernibles**:  $d(i, i) = 0$ : The distance of an object to itself is 0.
  - **Symmetry**:  $d(i, j) = d(j, i)$ : Distance is a symmetric function.
  - **Triangle inequality**:  $d(i, j) \leq d(i, k) + d(k, j)$ : Going directly from object  $i$  to object  $j$  in space is no more than making a detour over any other object  $k$ .
- A measure that satisfies these conditions is known as **metric**. Please note that the nonnegativity property is implied by the other three properties.





### 3. Similarity and Distance Measurements

#### 3.4. Dissimilarity of Numeric Data: Minkowski Distance

- **Example 2.20. Euclidean distance and Manhattan distance.** Let  $x_1 = (1, 2)$  and  $x_2 = (3, 5)$  represent two objects as shown in Fig. 2.10.
  - The Euclidean distance between the two is  $\sqrt{2^2 + 3^2} = 3.61$ .
  - The Manhattan distance between the two is  $2 + 3 = 5$ .



### 3. Similarity and Distance Measurements

#### 3.4. Dissimilarity of Numeric Data: Minkowski Distance

- **Minkowski distance** is a generalization of the Euclidean and Manhattan distances. It is defined as

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \dots + |x_{ip} - x_{jp}|^h}$$

where  $h$  is a real number such that  $h \geq 1$ . (Such a distance is also called  $L_p$  **norm** in some literature, where the symbol  $p$  refers to our notation of  $h$ ).

- $h = 1$ : represents the Manhattan distance ( $L_1$  norm)
- $h = 2$ : represents the Euclian distance ( $L_2$  norm)



### 3. Similarity and Distance Measurements

#### 3.4. Dissimilarity of Numeric Data: Minkowski Distance

- The **supremum distance** (also referred to as  $L_{max}$ ,  $L_{\infty}$  **norm**, and the **Chebyshev distance**) is a generalization of the Minkowski distance for  $h \rightarrow \infty$ . To compute it, we find the attribute  $f$  that gives the maximum difference in values between the two objects. This difference is the supremum distance, defined more formally as:

$$d(i, j) = \lim_{h \rightarrow \infty} \left( \sum_{f=1}^p |x_{if} - x_{jf}|^h \right)^{\frac{1}{h}} = \max_f |x_{if} - x_{jf}|$$

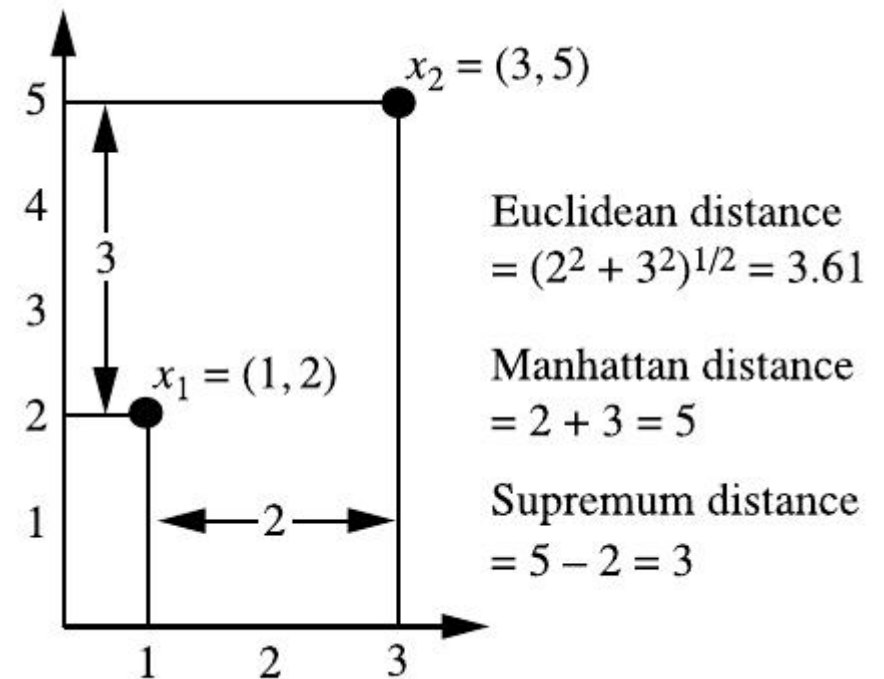
- The  $L_{\infty}$  norm is also known as the *uniform norm*.

### 3. Similarity and Distance Measurements

#### 3.4. Dissimilarity of Numeric Data: Minkowski Distance

- **Example 2.21. Supremum distance.**

Let's use the same two objects,  $x_1 = (1, 2)$  and  $x_2 = (3, 5)$ , as in Fig. 2.10. The second attribute gives the greatest difference between the values for the objects. That is,  $\max\{|3 - 1|, |5 - 2|\} = 3$ . This is the supremum distance between the two objects.





### 3. Similarity and Distance Measurements

#### 3.4. Dissimilarity of Numeric Data: Minkowski Distance

- If each attribute is assigned a weight according to its perceived importance, the **weighted Euclidean distance** can be computed as

$$d(i, j) = \sqrt{w_1(x_{i1} - x_{j1})^2 + w_2(x_{i2} - x_{j2})^2 + \dots + w_m(x_{ip} - x_{jp})^2}$$



## 3. Similarity and Distance Measurements

### 3.5. Proximity Measures for Ordinal Attributes

- The values of an ordinal attribute have a meaningful order or ranking about them, yet the magnitude between successive values is unknown. An example includes the sequence *small, medium, large* for a *size* attribute.
- Ordinal attributes may also be obtained from the discretization of numeric attributes by splitting the value range into a finite number of categories. These categories are organized into ranks. That is, the range of a numeric attribute can be mapped to an ordinal attribute  $f$  having  $M_f$  states.
  - For example, the range of the interval-scaled attribute temperature (in Celsius) can be organized into the following states:  $-30$  to  $-10$ ,  $-10$  to  $10$ , and  $10$  to  $30$ , representing the categories cold temperature, moderate temperature, and warm temperature, respectively.
  - Let  $M_f$  represent the number of possible states that an ordinal attribute can have. These ordered states define the ranking  $1, \dots, M_f$ .



## 3. Similarity and Distance Measurements

### 3.5. Proximity Measures for Ordinal Attributes

- “How are ordinal attributes handled?”
- The treatment of ordinal attributes is quite similar to that of numeric attributes when computing dissimilarity between objects. Suppose that  $f$  is an attribute from a set of ordinal attributes describing  $n$  objects. The dissimilarity computation with respect to  $f$  involves the following steps:
  - 1. The value of  $f$  for the  $i$ th object is  $x_{if}$ , and  $f$  has  $M_f$  ordered states, representing the ranking  $1, \dots, M_f$ . Replace each  $x_{if}$  by its corresponding rank,  $r_{if} \in \{1, \dots, M_f\}$ .
  - 2. Since each ordinal attribute can have a different number of states, it is often necessary to map the range of each attribute onto  $[0.0, 1.0]$  so that each attribute has equal weight. We perform such data normalization by replacing the rank  $r_{if}$  of the  $i$ th object in the  $f$ th attribute by
$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$
  - 3. Dissimilarity can then be computed using any of the distance measures for numeric attributes, using  $z_{if}$  to represent the  $f$  value for the  $i$ th object.

# 3. Similarity and Distance Measurements

## 3.5. Proximity Measures for Ordinal Attributes

- **Example 2.22. Dissimilarity between ordinal attributes.**

Suppose that we have the sample data shown earlier in Table 2.4, except that this time only the *object-identifier* and the continuous ordinal attribute, *test-2*, are available.

- There are three states for *test-2*: *fair*, *good*, and *excellent*, that is,  $M_f = 3$ .
- For step 1, if we replace each value for *test-2* by its rank, the four objects are assigned the ranks 3, 1, 2, and 3, respectively.
- Step 2 normalizes the ranking by mapping rank 1 to 0.0, rank 2 to 0.5, and rank 3 to 1.0.
- For step 3, we can use, say, the Euclidean distance, which results in the following dissimilarity matrix:

**Table 2.4** A sample data table containing attributes of mixed types.

| Object Identifier | Test-1 (nominal) | Test-2 (ordinal) | Test-3 (numeric) |
|-------------------|------------------|------------------|------------------|
| 1                 | code A           | excellent        | 45               |
| 2                 | code B           | fair             | 22               |
| 3                 | code C           | good             | 64               |
| 4                 | code A           | excellent        | 28               |

$$\begin{bmatrix} 0 & & & \\ 1.0 & 0 & & \\ 0.5 & 0.5 & 0 & \\ 0 & 1.0 & 0.5 & 0 \end{bmatrix}$$





## 3. Similarity and Distance Measurements

### 3.6. Dissimilarity for Attributes of Mixed Types

- In many real databases, objects are described by a mixture of attribute types.
- *“So, how can we compute the dissimilarity between objects of mixed attribute types?”*
  - Combines the different attributes into a single dissimilarity matrix, bringing all of the meaningful attributes onto a common scale of the interval  $[0.0, 1.0]$ .

### 3. Similarity and Distance Measurements

#### 3.6. Dissimilarity for Attributes of Mixed Types

- Suppose that the data set contains  $p$  attributes of mixed types. The dissimilarity  $d(i, j)$  between objects  $i$  and  $j$  is defined as

$$d(i, j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}}$$

where the indicator  $\delta_{ij}^{(f)} = 0$  if either

nếu thiếu 1 trong 2 thành phần

(1)  $x_{if}$  or  $x_{jf}$  is missing (i.e., there is no measurement of attribute  $f$  for object  $i$  or object  $j$ ), or

(2)  $x_{if} = x_{jf} = 0$  and attribute  $f$  is asymmetric binary;

(3) otherwise,  $\delta_{ij}^{(f)} = 1$ .

### 3. Similarity and Distance Measurements

#### 3.6. Dissimilarity for Attributes of Mixed Types

- The contribution of attribute  $f$  to the dissimilarity between  $i$  and  $j$  (i.e.,  $d_{ij}^{(f)}$ ) is computed dependent on its type:
  - If  $f$  is numeric:  $d_{ij}^{(f)} = \frac{|x_{if} - x_{jf}|}{\max_f - \min_f}$ , where  $\max_f$  and  $\min_f$  are the maximum and minimum values of attribute  $f$ , respectively;
  - If  $f$  is nominal or binary:  $d_{ij}^{(f)} = 0$  if  $x_{if} = x_{jf}$ ; otherwise,  $d_{ij}^{(f)} = 1$ ; and
  - If  $f$  is ordinal: compute the ranks  $r_{if}$  and  $z_{if} = \frac{r_{if} - 1}{M_{if} - 1}$ , and treat  $z_{if}$  as numeric.

### 3. Similarity and Distance Measurements

#### 3.6. Dissimilarity for Attributes of Mixed Types

- Example 2.23. Dissimilarity between attributes of mixed types. Now we will consider all of the attributes, which are of different types.

**Table 2.4** A sample data table containing attributes of mixed types.

| Object Identifier | Test-1 (nominal) | Test-2 (ordinal) | Test-3 (numeric) |
|-------------------|------------------|------------------|------------------|
| 1                 | code A           | excellent        | 45               |
| 2                 | code B           | fair             | 22               |
| 3                 | code C           | good             | 64               |
| 4                 | code A           | excellent        | 28               |

– Dissimilarity matrix for test-1 (Example 2.18)

$$\begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Dissimilarity matrix for test-2 (Example 2.22)

$$\begin{bmatrix} 0 & & & \\ 1.0 & 0 & & \\ 0.5 & 0.5 & 0 & \\ 0 & 1.0 & 0.5 & 0 \end{bmatrix}$$

### 3. Similarity and Distance Measurements

#### 3.6. Dissimilarity for Attributes of Mixed Types

- We need to compute the dissimilarity matrix for the third attribute, *test-3* (which is numeric). That is, we must compute  $d_{ij}^{(3)}$ 
  - $\max_{test-3} = 64$  and  $\min_{test-3} = 22$ .
  - The resulting dissimilarity matrix for *test-3* is

$$\begin{bmatrix} 0 & & & \\ 0.55 & 0 & & \\ 0.45 & 1.00 & 0 & \\ 0.40 & 0.14 & 0.86 & 0 \end{bmatrix}$$

### 3. Similarity and Distance Measurements

#### 3.6. Dissimilarity for Attributes of Mixed Types

- The indicator  $\delta_{ij}^{(f)} = 1$  for each of the three attributes,  $f$ .

$$\begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & & & \\ 1.0 & 0 & & \\ 0.5 & 0.5 & 0 & \\ 0 & 1.0 & 0.5 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & & & \\ 0.55 & 0 & & \\ 0.45 & 1.00 & 0 & \\ 0.40 & 0.14 & 0.86 & 0 \end{bmatrix}$$

- The resulting dissimilarity matrix obtained for the data described by the three attributes of mixed types is:

$$\begin{bmatrix} 0 & & & \\ 0.85 & 0 & & \\ 0.65 & 0.83 & 0 & \\ 0.13 & 0.71 & 0.79 & 0 \end{bmatrix}$$

# 3. Similarity and Distance Measurements

## 3.7. Cosine Similarity

- **Cosine similarity** measures the similarity between two vectors of an inner product space. It is measured by the cosine of the angle between two vectors and determines whether two vectors are pointing in roughly the same direction. It is often used to measure document similarity in text analysis.
- A document can be represented by thousands of attributes, each recording the frequency of a particular word (such as a keyword) or phrase in the document. Thus each document is an object represented by what is called a *term-frequency vector*.
  - Cosine similarity is a measure of similarity that can be used to compare documents or, say, give a ranking of documents with respect to a given vector of query words.

**Table 2.7 Document vector or term-frequency vector.**

| Document  | Team | Coach | Hockey | Baseball | Soccer | Penalty | Score | Win | Loss | Season |
|-----------|------|-------|--------|----------|--------|---------|-------|-----|------|--------|
| Document1 | 5    | 0     | 3      | 0        | 2      | 0       | 0     | 2   | 0    | 0      |
| Document2 | 3    | 0     | 2      | 0        | 1      | 1       | 0     | 1   | 0    | 1      |
| Document3 | 0    | 7     | 0      | 2        | 1      | 0       | 0     | 3   | 0    | 0      |
| Document4 | 0    | 1     | 0      | 0        | 1      | 2       | 2     | 0   | 3    | 0      |



## 3. Similarity and Distance Measurements

### 3.7. Cosine Similarity

- Let  $\mathbf{x}$  and  $\mathbf{y}$  be two vectors for comparison. Using the cosine measure as a similarity function, we have

$$\text{sim}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

- where  $\|\mathbf{x}\|$  is the Euclidean norm of vector  $\mathbf{x} = (x_1, x_2, \dots, x_p)$ , defined as  $\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_p^2}$ . Conceptually, it is the length of the vector. Similarly,  $\|\mathbf{y}\|$  is the Euclidean norm of vector  $\mathbf{y}$ .
- The measure computes the cosine of the angle between vectors  $\mathbf{x}$  and  $\mathbf{y}$ .
  - A cosine value of 0 means that the two vectors are at 90 degrees to each other (orthogonal) and have no match. The closer the cosine value to 1, the smaller the angle and the greater the match between vectors.
- Note that because the cosine similarity measure does not obey all of the properties of Section 2.3.4 defining metric measures, it is referred to as a *nonmetric measure*.



## 3. Similarity and Distance Measurements

### 3.7. Cosine Similarity

- Example 2.24. Cosine similarity between two term-frequency vectors.** Suppose that  $\mathbf{x}$  and  $\mathbf{y}$  are the first two term-frequency vectors in Table 2.7.

| Table 2.7 Document vector or term-frequency vector. |      |       |        |          |        |         |       |     |      |        |
|-----------------------------------------------------|------|-------|--------|----------|--------|---------|-------|-----|------|--------|
| Document                                            | Team | Coach | Hockey | Baseball | Soccer | Penalty | Score | Win | Loss | Season |
| <i>Document1</i>                                    | 5    | 0     | 3      | 0        | 2      | 0       | 0     | 2   | 0    | 0      |
| <i>Document2</i>                                    | 3    | 0     | 2      | 0        | 1      | 1       | 0     | 1   | 0    | 1      |
| <i>Document3</i>                                    | 0    | 7     | 0      | 2        | 1      | 0       | 0     | 3   | 0    | 0      |
| <i>Document4</i>                                    | 0    | 1     | 0      | 0        | 1      | 2       | 2     | 0   | 3    | 0      |

$$\begin{aligned}\mathbf{x} \cdot \mathbf{y} &= 5 \times 3 + 0 \times 0 + 3 \times 2 + 0 \times 0 + 2 \times 1 + 0 \times 1 + 0 \times 0 + 2 \times 1 \\ &\quad + 0 \times 0 + 0 \times 1 = 25\end{aligned}$$

$$\|\mathbf{x}\| = \sqrt{5^2 + 0^2 + 3^2 + 0^2 + 2^2 + 0^2 + 0^2 + 2^2 + 0^2 + 0^2} = 6.48$$

$$\|\mathbf{y}\| = \sqrt{3^2 + 0^2 + 2^2 + 0^2 + 1^2 + 1^2 + 0^2 + 1^2 + 0^2 + 1^2} = 4.12$$

$$\text{sim}(\mathbf{x}, \mathbf{y}) = 0.94.$$

## 3. Similarity and Distance Measurements

### 3.8. Measuring Similar Distributions: the Kullback-Leibler Divergence (KL Divergence)

- *Kullback-Leibler divergence* is a measure that has been popularly used in the data mining literature to measure the difference between two probability distributions over the same variable  $x$ .
- This concept was originated in probability theory and information theory.
- The KL divergence, which is closely related to *relative entropy*, *information divergence*, and *information for discrimination*, is a nonsymmetric measure of the difference between two probability distributions  $p(x)$  and  $q(x)$ .
- Specifically, the KL divergence of  $q(x)$  from  $p(x)$ , denoted  $D_{KL}(p(x)||q(x))$ , is a measure of the information loss when  $q(x)$  is used to approximate  $p(x)$ .
- Let  $p(x)$  and  $q(x)$  be two probability distributions of a discrete random variable  $x$ .
  - That is, both  $p(x)$  and  $q(x)$  sum up to 1, and  $p(x) > 0$  and  $q(x) > 0$  for any  $x$  in  $X$ .

$$D_{KL}(p(x)||q(x)) = \sum_{x \in X} p(x) \ln \frac{p(x)}{q(x)}$$

### 3. Similarity and Distance Measurements

#### 3.8. Measuring Similar Distributions: the Kullback-Leibler Divergence (KL Divergence)

- The continuous version of the KL divergence is

$$D_{KL}(p(x)||q(x)) = \int_{-\infty}^{\infty} p(x) \ln \frac{p(x)}{q(x)} dx$$

- Although the KL divergence measures the “distance” between two distributions, it is not a distance measure. This is because that the KL divergence is not a metric measure.
- It is not symmetric: the KL from  $p(x)$  to  $q(x)$  is generally not the same as the KL from  $q(x)$  to  $p(x)$ .
- Furthermore, it need not satisfy triangular inequality.
- Nevertheless,  $D_{KL}(p(x)||q(x))$  is a nonnegative measure.
  - $D_{KL}(p(x)||q(x)) \geq 0$  and  $D_{KL}(p(x)||q(x)) = 0$  if and only if  $p(x) = q(x)$ .

## 3. Similarity and Distance Measurements

### 3.8. Measuring Similar Distributions: the Kullback-Leibler Divergence (KL Divergence)

- **Example 2.25. Computing the KL divergence by smoothing.** Suppose there are two sample distributions  $P$  and  $Q$  as follows:
  - $P : (a : 3/5, b : 1/5, c : 1/5)$
  - $Q : (a : 5/9, b : 3/9, d : 1/9)$ .
- To compute the KL divergence  $D_{KL}(P||Q)$ , we introduce a small constant  $\varepsilon$ , for example  $\varepsilon = 10^{-3}$ , and define a smoothed version of  $P$  and  $Q$ ,  $P'$  and  $Q'$ , as follows.
- The sample set observed in  $P$ ,  $SP = \{a, b, c\}$ . Similarly,  $SQ = \{a, b, d\}$ . The union set is  $SU = \{a, b, c, d\}$ . By smoothing, the missing symbols can be added to each distribution accordingly, with the small probability  $\varepsilon$ .
- Thus we have  $P' : (a : 3/5 - \varepsilon/3, b : 1/5 - \varepsilon/3, c : 1/5 - \varepsilon/3, d : \varepsilon)$  and  $Q' : (a : 5/9 - \varepsilon/3, b : 3/9 - \varepsilon/3, c : \varepsilon, d : 1/9 - \varepsilon/3)$ .  $D_{KL}(P', Q')$  can be computed easily.



## 4. Data Quality, Data Cleaning, and Data Integration

### 4.1. Data Quality Measures

- Data have quality if they satisfy the requirements of the intended use.
  - There are many factors comprising **data quality**, including *accuracy*, *completeness*, *consistency*, *timeliness*, *believability*, and *interpretability*.
- Inaccuracy or noisy
  - Containing errors, or values that deviate from expected
- Incomplete
  - Lacking attribute values or certain attributes of interest, or containing only aggregate data
- Inconsistent
  - Containing discrepancies,



## 4. Data Quality, Data Cleaning, and Data Integration

### 4.1. Data Quality Measures

- Timeliness
  - The fact that the data are not updated in a timely fashion has a negative impact on the data quality. However, once all of the data are received, it is correct.
- Believability
  - reflects how much the data are trusted by users
- Interpretability
  - reflects how easily the data are understood.



## 4. Data Quality, Data Cleaning, and Data Integration

### 4.2. Data Cleaning

- Real-world data tend to be incomplete, noisy, and inconsistent.
- *Data cleaning* (or *data cleansing*) routines attempt to fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies in the data.
  - Handling missing values
  - Data smoothing techniques
  - Data cleaning as a process

# 4. Data Quality, Data Cleaning, and Data Integration

## 4.2. Data Cleaning

### Missing Values

- Note that many tuples have no recorded value for several attributes.
- How can you go about filling in the missing values for an attribute?
  - **1. Ignore the tuple**
    - It is especially poor when the percentage of missing values per attribute varies considerably. By ignoring the tuple, we do not make use of the remaining attributes' values in the tuple. Such data could have been useful to the task at hand.
  - **2. Fill in the missing value manually**
    - In general, this approach is time consuming and may not be feasible given a large data set with many missing values.
  - **3. Use a global constant to fill in the missing value**
    - Replace all missing attribute values by the same constant such as a label like “*Unknown*” or  $-\infty$ . If missing values are replaced by, say, “*Unknown*”, then the mining program may mistakenly think that they form an interesting concept, since they all have a value in common—that of “*Unknown*.” Hence, although this method is simple, it is not foolproof.



# 4. Data Quality, Data Cleaning, and Data Integration

## 4.2. Data Cleaning

### Missing Values

- **4. Use a measure of central tendency for the attribute (e.g., the mean or median) to fill in the missing value**
  - For normal (symmetric) data distributions, the mean can be used, whereas skewed data distribution should employ the median.
- **5. Use the attribute mean or median for all samples belonging to the same class as the given tuple**
  - For example, if classifying customers according to *credit\_risk*, we may replace the missing value with the mean income value for customers in the same credit risk category as that of the given tuple.
  - If the data distribution for a given class is skewed, the median value is a better choice.
- **6. Use the most probable value to fill in the missing value**
  - This may be determined with regression, inference-based tools using a Bayesian formalism or decision tree induction.

# 4. Data Quality, Data Cleaning, and Data Integration

## 4.2. Data Cleaning

### Noisy Data

- *What is noise?*
- **Noise** is a random error or variance in a measured variable. Given a numeric attribute such as, say, *price*, how can we “smooth” out the data to remove the noise?
- **Binning**
  - Binning methods smooth a sorted data value by consulting its “neighborhood,” that is, the values around it. The sorted values are distributed into a number of “buckets,” or bins.
  - Because binning methods consult the neighborhood of values, they perform *local* smoothing.

# 4. Data Quality, Data Cleaning, and Data Integration

## 4.2. Data Cleaning

### Noisy Data

Sorted data for *price* (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34

#### Partition into (equal-frequency) bins:

Bin 1: 4, 8, 15

Bin 2: 21, 21, 24

Bin 3: 25, 28, 34

#### Smoothing by bin means:

Bin 1: 9, 9, 9

Bin 2: 22, 22, 22

Bin 3: 29, 29, 29

#### Smoothing by bin boundaries:

Bin 1: 4, 4, 15

Bin 2: 21, 21, 24

Bin 3: 25, 25, 34

- In this example, the data for *price* are first sorted and then partitioned into *equal-frequency* bins of size 3 (i.e., each bin contains three values).
- In **smoothing by bin means**, each value in a bin is replaced by the mean value of the bin.
  - For example, the mean of the values 4, 8, and 15 in Bin 1 is 9. Therefore each original value in this bin is replaced by the value 9.
  - Similarly, **smoothing by bin medians** can be employed, in which each bin value is replaced by the bin median.
- In **smoothing by bin boundaries**, the minimum and maximum values in a given bin are identified as the *bin boundaries*. Each bin value is then replaced by the closest boundary value.
  - In general, the larger the width, the greater the effect of the smoothing. Alternatively, bins may be *equal width*, where the interval range of values in each bin is constant. Binning is also used as a discretization technique.

# 4. Data Quality, Data Cleaning, and Data Integration

## 4.2. Data Cleaning

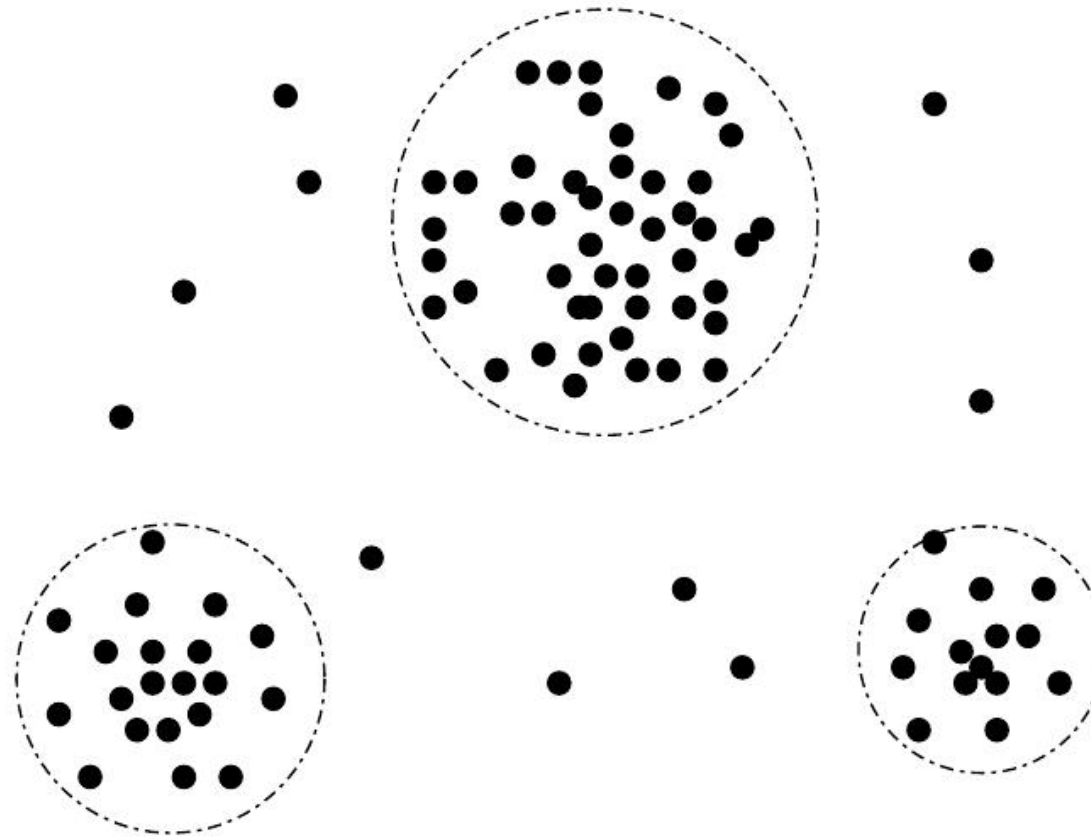
### Noisy Data

- **Regression:** Data smoothing can also be done by regression, a technique that conforms data values to a function.
  - Linear regression involves finding the “best” line to fit two attributes (or variables) so that one attribute can be used to predict the other.
  - Multiple linear regression is an extension of linear regression, where more than two attributes are involved, and the data are fit to a multidimensional surface.
- **Outlier analysis:** Outliers may be detected by clustering, for example, where similar values are organized into groups or “clusters.” Intuitively, values that fall outside of the set of clusters may be considered as outliers (Fig. 2.12).

# 4. Data Quality, Data Cleaning, and Data Integration

## 4.2. Data Cleaning

### Noisy Data



**FIGURE 2.12**

A 2-D customer data plot with respect to customer locations in a city, showing three data clusters. Outliers may be detected as values that fall outside of the cluster sets.

## 4. Data Quality, Data Cleaning, and Data Integration

### 4.2. Data Cleaning

#### Data Cleaning as a Process

- Data cleaning is usually performed as an iterative two-step process consisting of discrepancy detection and data transformation.
- The first step in data cleaning as a process is *discrepancy detection*.
- Once we find discrepancies, we typically need to define and apply (a series of) transformations to correct them.

## 4. Data Quality, Data Cleaning, and Data Integration

### 4.2. Data Cleaning

#### Data Integration

- Data mining often requires data integration - the merging of data from multiple data stores. Careful integration can help reduce and avoid redundancies and inconsistencies in the resulting data set.
- The semantic heterogeneity and structure of data pose great challenges in data integration.

# 4. Data Quality, Data Cleaning, and Data Integration

## 4.2. Data Cleaning

### Data Integration

- **Entity Identification Problem**

- There are a number of issues to consider during data integration. *Schema integration* and *object matching* can be tricky. How can equivalent real-world entities from multiple data sources be matched up? This is referred to as the **entity identification problem**.
- When matching attributes from one database to another during integration, special attention must be paid to the structure of the data.



# 4. Data Quality, Data Cleaning, and Data Integration

## 4.2. Data Cleaning

### Data Integration

- **Redundancy and Correlation Analysis**

- *Redundancy* is another important issue in data integration. An attribute (such as annual revenue, for instance) may be redundant if it can be “derived” from another attribute or set of attributes.
- Inconsistencies in attribute or dimension naming can also cause redundancies in the resulting data set.
- Some redundancies can be detected by **correlation analysis**. Given two attributes, such analysis can measure how strongly one attribute implies the other, based on the available data.
  - For nominal data, we can use the  $\chi^2$  (chi-square) test.
  - For numeric attributes, we can use the correlation coefficient and covariance, both of which assess how one attribute’s values vary from those of another.

# 4. Data Quality, Data Cleaning, and Data Integration

## 4.2. Data Cleaning

### Data Integration

- **Tuple Duplication**

- In addition to detecting redundancies between attributes, duplication should also be detected at the tuple level (e.g., where there are two or more identical tuples for a given unique data entry).
- The use of denormalized tables (often done to improve performance by avoiding joins) is another source of data redundancy.
  - Inconsistencies often arise between various duplicates, due to inaccurate data entry or updating some but not all data occurrences.
  - For example, if a purchase order database contains attributes for the purchaser's name and address instead of a key to this information in a purchaser database, discrepancies can occur, such as the same purchaser's name appearing with different addresses within the purchase order database.

# 4. Data Quality, Data Cleaning, and Data Integration

## 4.2. Data Cleaning

### Data Integration

- **Data Value Conflict Detection and Resolution**

- For the same real-world entity, attribute values from different sources may differ. This may be due to differences in representation, scaling, or encoding.
  - For instance, a *weight* attribute may be stored in metric units in one system and British imperial units in another.
  - For a hotel chain, the *price* of rooms in different cities may involve not only different currencies but also different services (e.g., free breakfast) and taxes.
  - When exchanging information between schools, for example, each school may have its own curriculum and grading scheme. One university may adopt a quarter system, offer three courses on database systems, and assign grades from A+ to F, whereas another may adopt a semester system, offer two courses on databases, and assign grades from 1 to 10. It is difficult to work out precise course-to-grade transformation rules between the two universities, making information exchange difficult.
- Attributes may also differ on the abstraction level, where an attribute in one system is recorded at, say, a lower abstraction level than the “same” attribute in another.
  - For example, the `total_sales` in one database may refer to one branch of the company, whereas an attribute of the same name in another database may refer to the total sales for the stores in a given region.



## 5. Data Transformation

- In *data transformation*, the data are transformed or consolidated into forms appropriate for mining.
  - The resulting mining process may be more efficient, and the patterns found may be easier to understand.
- Various strategies for data transformation have been developed.
  - Normalization
  - Discretization
  - Data compression
  - Sampling



## 5. Data Transformation

### 5.1. Normalization

- The measurement unit used can affect the data analysis.
  - For example, changing measurement units from meters to inches for *height*, or from kilograms to pounds for *weight*, may lead to very different results.
  - In general, expressing an attribute in smaller units will lead to a larger range for that attribute and thus tend to give such an attribute greater effect or “weight”.
  - To help avoid dependence on the choice of measurement units, the data should be *normalized* or *standardized*. This involves transforming the data to fall within a smaller or common range such as  $[-1.0, 1.0]$  or  $[0.0, 1.0]$ .
- Normalizing the data attempts to give all attributes an equal weight.
- There are many methods for data normalization: min-max normalization, z-score normalization, and normalization by decimal scaling.

## 5. Data Transformation

### 5.1. Normalization

#### Min-max Normalization

- Let  $A$  be a numeric attribute with  $n$  observed values,  $v_1, v_2, \dots, v_n$ .
- **Min-max normalization** performs a linear transformation on the original data.
- Suppose that  $\min_A$  and  $\max_A$  are the minimum and maximum values of an attribute,  $A$ . Min-max normalization maps a value,  $v_i$ , of  $A$  to  $v_i'$  in the range  $[\text{new\_min}_A, \text{new\_max}_A]$  by computing

$$v_i' = \frac{v_i - \min_A}{\max_A - \min_A} (\text{new\_max}_A - \text{new\_min}_A) + \text{new\_min}_A$$

- Min-max normalization preserves the relationships among the original data values. It will encounter an “out-of-bounds” error if a future input case for normalization falls outside of the original data range for  $A$ .

## 5. Data Transformation

### 5.1. Normalization

#### Min-max Normalization

- **Example 2.26. Min-max normalization.** Suppose that the minimum and maximum values for the attribute *income* are \$12,000 and \$98,000, respectively. We would like to map income to the range [0.0, 1.0].
- By min-max normalization, a value of \$73,600 for income is transformed to  $(73,600 - 12,000) / (98,000 - 12,000) (1.0 - 0) + 0 = 0.716$ .

## 5. Data Transformation

### 5.1. Normalization

#### Z-Score Normalization (zero-mean normalization)

- The values for an attribute,  $A$ , are normalized based on the mean (i.e., average) and standard deviation of  $A$ . A value,  $v_i$ , of  $A$  is normalized to  $v_i'$  by computing

$$v_i' = \frac{v_i - \bar{A}}{\sigma_A}$$

where  $\bar{A}$  and  $\sigma_A$  are the mean and standard deviation, respectively, of attribute  $A$ .

- This method of normalization is useful when the actual minimum and maximum of attribute  $A$  are unknown or when there are outliers that dominate the min-max normalization.
- Use Z-Scores to identifying Outliers:
  - Z-scores can help identify outliers by flagging data points that are far from the mean. Typically, values with z-scores beyond a certain threshold (often considered as  $\pm 3.0$  or  $\pm 2.5$ ) are considered outliers.



## 5. Data Transformation

### 5.1. Normalization

#### Z-Score Normalization (zero-mean normalization)

- **Example 2.27. z-score normalization.** Suppose that the mean and standard deviation of the values for the attribute *income* are \$54,000 and \$16,000, respectively. With z-score normalization, a value of \$73,600 for income is transformed to  $(73,600 - 54,000) / 16,000 = 1.225$ .

- A variation of this z-score normalization replaces the standard deviation by the mean absolute deviation of  $A$ . The mean absolute deviation of  $A$ , denoted  $s_A$ , is

$$s_A = \frac{1}{n} (|v_1 - \bar{A}| + |v_2 - \bar{A}| + \dots + |v_n - \bar{A}|)$$

- Thus z-score normalization using the mean absolute deviation is

$$v'_i = \frac{v_i - \bar{A}}{s_A}$$

- The mean absolute deviation,  $s_A$ , is more robust to outliers than the standard deviation,  $\sigma_A$ . When computing the mean absolute deviation, the deviations from the mean (i.e.,  $|x_i - \bar{x}|$ ) are not squared; hence, the effect of outliers is somewhat reduced.

## 5. Data Transformation

### 5.1. Normalization

#### Normalization by Decimal Scaling

- **Normalization by decimal scaling** normalizes by moving the decimal point of values of attribute  $A$ . The number of decimal points moved depends on the maximum absolute value of  $A$ . A value,  $v_i$ , of  $A$  is normalized to  $v_i'$  by computing

$$v_i' = \frac{v_i}{10^j}$$

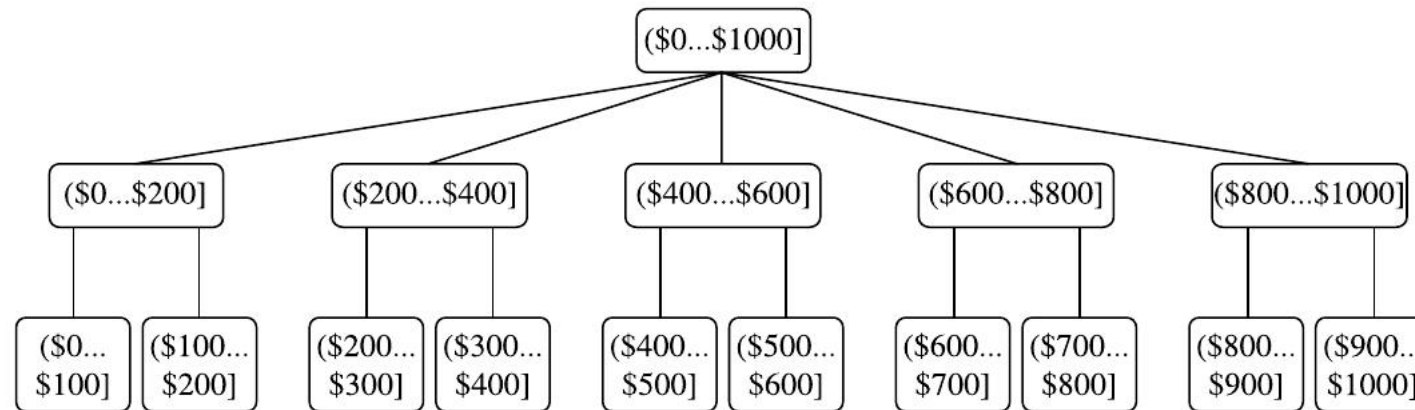
where  $j$  is the smallest integer such that  $\max(|v_i'|) < 1$ .

- **Example 2.28. Decimal scaling.** Suppose that the recorded values of  $A$  range from  $-986$  to  $917$ . The maximum absolute value of  $A$  is  $986$ . To normalize by decimal scaling, we therefore divide each value by  $1000$  (i.e.,  $j = 3$ ) so that  $-986$  normalizes to  $-0.986$  and  $917$  normalizes to  $0.917$ .

## 5. Data Transformation

### 5.2. Discretization

- Data discretization is a common data transformation technique, where the raw values of a numeric attribute (e.g., *age*) are replaced by interval labels (e.g., 0–10, 11–20, etc.) or conceptual labels (e.g., *youth*, *adult*, *senior*). The labels, in turn, can be recursively organized into higher-level concepts, resulting in a concept hierarchy for the numeric attribute.



**FIGURE 2.13**

A concept hierarchy for the attribute *price*, where an interval  $(\$X \dots \$Y]$  denotes the range from  $\$X$  (exclusive) to  $\$Y$  (inclusive).



## 5. Data Transformation

### 5.2. Discretization

- Discretization techniques can be categorized based on how the discretization is performed, such as whether it uses class information or which direction it proceeds (i.e., top-down vs. bottom-up).
- If the discretization process uses class information, then we say it is *supervised discretization*. Otherwise, it is *unsupervised*.

# 5. Data Transformation

## 5.2. Discretization

### Discretization by Binning

- Binning is a top-down splitting technique based on a specified number of bins.
- Binning methods are used for data smoothing: smoothing by bin means, smoothing by bin medians, smoothing by bin boundaries.
- These methods are also used as discretization methods for data reduction and concept hierarchy generation.
- For example, attribute values can be discretized by applying equal-width or equal-frequency binning and then replacing each bin value by the bin mean or median, as in smoothing by bin means or smoothing by bin medians, respectively.
- These techniques can be applied recursively to the resulting partitions to generate concept hierarchies.
- Binning does not use class information and is therefore an unsupervised discretization technique. It is sensitive to the user-specified number of bins, as well as the presence of outliers.

## 5. Data Transformation

### 5.2. Discretization

#### Discretization by Histogram Analysis

- Histogram analysis is an unsupervised discretization technique because it does not use class information.
- A histogram partitions the values of an attribute,  $A$ , into disjoint ranges called *buckets* or *bins*.
- If each bucket represents only a single attribute-value/frequency pair, the buckets are called *singleton buckets*. Singleton buckets are useful for storing high-frequency outliers. Often, buckets instead represent continuous ranges for the given attribute.

## 5. Data Transformation

### 5.2. Discretization

#### Discretization by Histogram Analysis

- **Example 2.29.** The following data are a list of prices for commonly sold items in the company (rounded to the nearest dollar). The numbers have been sorted: 1, 1, 5, 5, 5, 5, 5, 8, 8, 10, 10, 10, 10, 12, 14, 14, 14, 15, 15, 15, 15, 15, 15, 18, 18, 18, 18, 18, 18, 20, 20, 20, 20, 20, 20, 20, 21, 21, 21, 21, 25, 25, 25, 25, 25, 28, 28, 30, 30, 30.

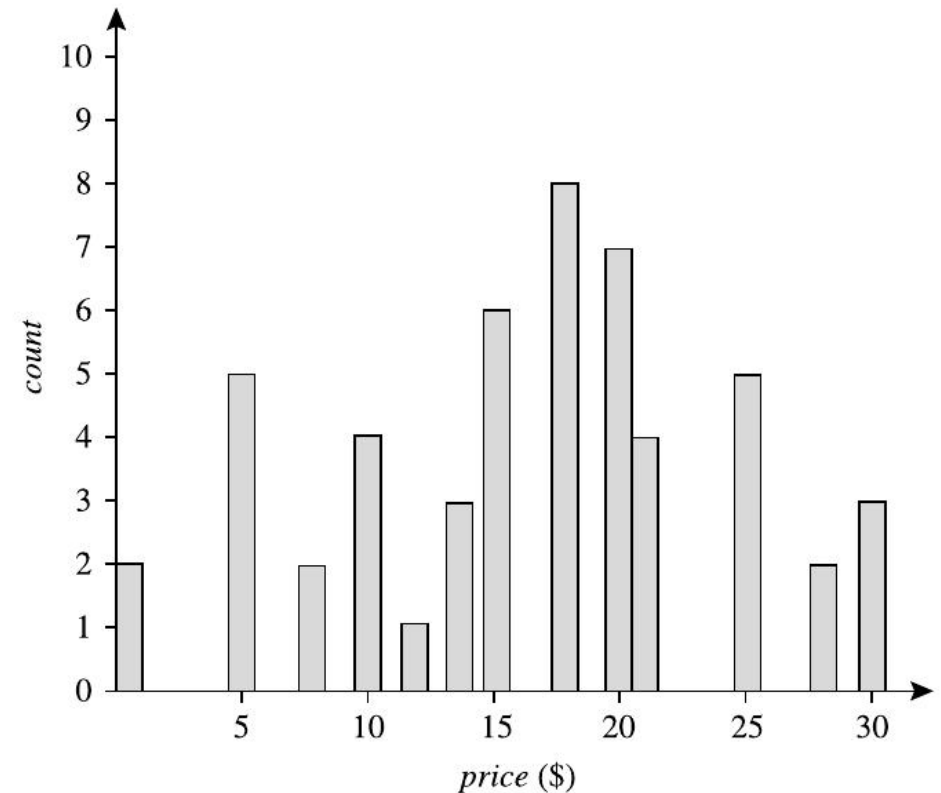


FIGURE 2.14

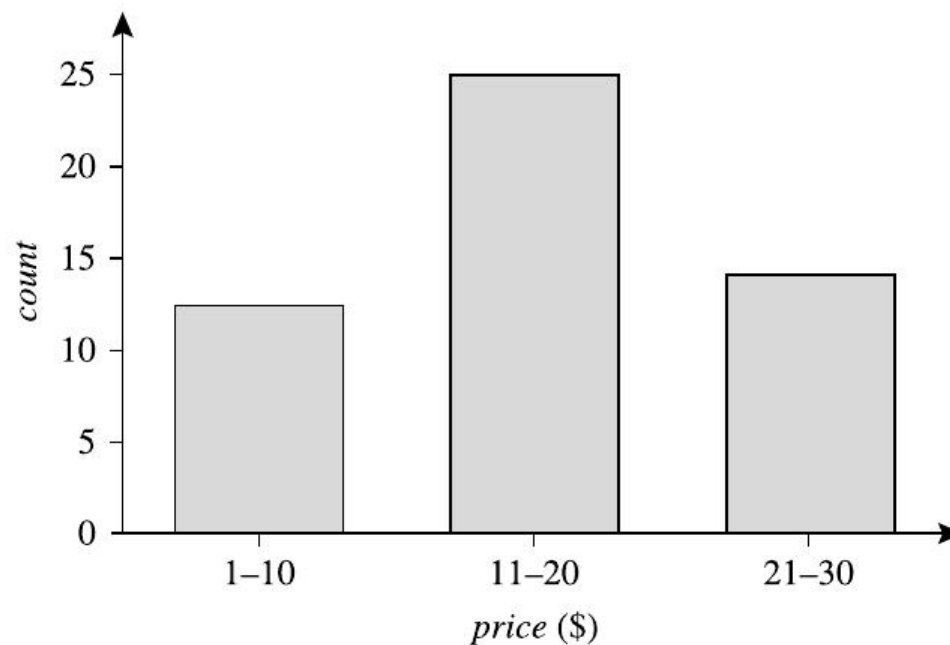
A histogram for *price* using singleton buckets—each bucket represents one price–value/frequency pair.

## 5. Data Transformation

### 5.2. Discretization

#### Discretization by Histogram Analysis

- To further reduce the data, it is common to have each bucket denote a continuous value range for the given attribute.



**FIGURE 2.15**

An equal-width histogram for *price*, where values are aggregated so that each bucket has a uniform width of \$10.



## 5. Data Transformation

### 5.2. Discretization

#### Discretization by Histogram Analysis

- “*How are the buckets determined and the attribute values partitioned?*”
- There are several partitioning rules, including the following:
  - **Equal-width:** In an equal-width histogram, the width of each bucket range is uniform (e.g., the width of \$10 for the buckets in Fig. 2.15).
  - **Equal-frequency** (or equal-depth): In an equal-frequency histogram, the buckets are created so that, roughly, the frequency of each bucket is constant (i.e., each bucket contains roughly the same number of contiguous data samples).



## 5. Data Transformation

### 5.3. Data Compression

- In data compression, transformations are applied so as to obtain a reduced or “compressed” representation of the original data.
- If the original data can be *reconstructed* from the compressed data without any information loss, the data reduction is called **lossless**.
- If, instead, we can reconstruct only an approximation of the original data, then the data reduction is called **lossy**.
- There are several lossless algorithms for string compression; however, they typically allow only limited data manipulation.
- Dimensionality reduction techniques can also be considered as forms of data compression.



## 5. Data Transformation

### 5.3. Data Compression

- The **discrete wavelet transform (DWT)** is a linear signal processing technique that, when applied to a data vector  $\mathbf{x}$ , transforms it to a numerically different vector,  $\mathbf{x}'$ , of **wavelet coefficients**. The two vectors are of the same length.
- *“How can this technique be useful for data reduction if the wavelet transformed data are of the same length as the original data?”*
  - The usefulness lies in the fact that the wavelet transformed data can be truncated. A compressed approximation of the data can be retained by storing only a small fraction of the strongest wavelet coefficients.
  - For example, all wavelet coefficients larger than some user-specified threshold can be retained. All other coefficients are set to 0.
    - The resulting data representation is therefore very sparse, so that operations that can take advantage of data sparsity are computationally very fast if performed in wavelet space.
    - The technique also works to remove noise without smoothing out the main features of the data, making it effective for data cleaning as well.
- Given a set of coefficients, an approximation of the original data can be constructed by applying the *inverse* of the DWT used.



## 5. Data Transformation

### 5.3. Data Compression

- The DWT is closely related to the *discrete Fourier transform (DFT)*, a signal processing technique involving sines and cosines.
- In general, however, the DWT achieves better lossy compression.
  - That is, if the same number of coefficients is retained for a DWT and a DFT of a given data vector, the DWT version will often provide a more accurate approximation of the original data.
  - Hence, for an equivalent approximation, the DWT requires less space than the DFT. Unlike the DFT, wavelets are quite localized in space, contributing to the conservation of local detail.
- There is only one DFT, yet there are several families of DWTs.
- Wavelet transforms give good results on sparse or skewed data and on data with ordered attributes. Lossy compression by wavelets is reportedly better than JPEG compression, the current commercial standard. Wavelet transforms have many real-world applications, including the compression of fingerprint images, computer vision, analysis of time-series data, and data cleaning.



## 5. Data Transformation

### 5.3. Data Compression

- Popular wavelet transforms include the Haar-2, Daubechies-4, and Daubechies-6. The general procedure for applying a discrete wavelet transform uses a hierarchical *pyramid algorithm* that halves the data at each iteration, resulting in fast computational speed.
- The method is as follows:
  1. The length,  $L$ , of the input data vector must be an integer power of 2. This condition can be met by padding the data vector with zeros as necessary ( $L \geq n$ ).
  2. Each transform involves applying two functions. The first applies some data smoothing, such as sum or weighted average. The second performs a weighted difference, which acts to bring out the detailed features of the data.
  3. The two functions are applied to pairs of data points in  $X$ , that is, to all pairs of measurements  $(x_{2i}, x_{2i+1})$ . This results in two data sets of length  $L/2$ . In general, these represent a smoothed or low-frequency version of the input data (approximation coefficients) and the high-frequency content of it (detail coefficients), respectively.
  4. The two functions are recursively applied to the data sets obtained in the previous iteration, until the resulting data sets obtained are of length 2.
  5. Selected values from the data sets obtained in the previous iterations are designated as the wavelet coefficients of the transformed data.

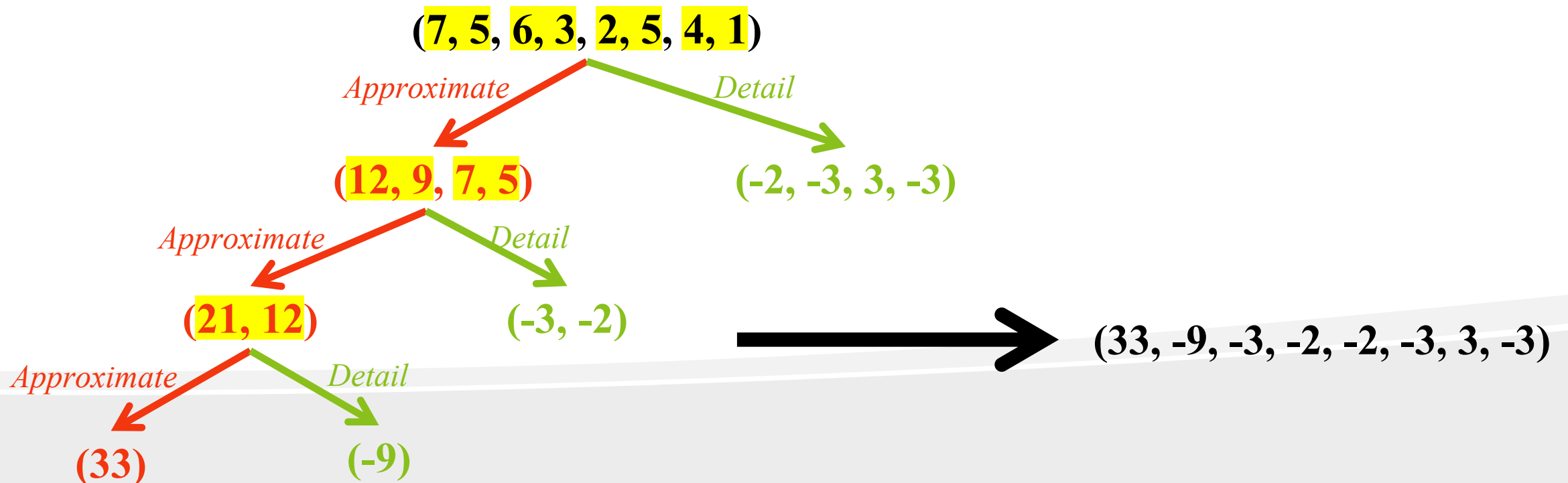
## 5. Data Transformation

### 5.3. Data Compression

- The 1D Haar Wavelet Transformation

- Approximation** =  $x_{2i} + x_{2i+1}$       **Detail** =  $x_{2i+1} - x_{2i}$

- $X = (7, 5, 6, 3, 2, 5, 4, 1)$        $n = 8$



## 5. Data Transformation

### 5.3. Data Compression

- Matrix of Haar Wavelet Transformation

- $$X = \begin{pmatrix} 7 \\ 5 \\ 6 \\ 3 \\ 2 \\ 5 \\ 4 \\ 1 \end{pmatrix} \quad W = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 & 1 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 \end{pmatrix}$$

- $W.X$



## 5. Data Transformation

### 5.4. Sampling

- Sampling can be used as a data reduction technique because it allows a large data set to be represented by a much smaller random data sample (or subset).
- Suppose that a large data set,  $D$ , contains  $N$  tuples.
- **Simple random sample without replacement (SRSWOR) of size  $s$ :** This is created by drawing  $s$  samples from  $D$ , and every time a sample is drawn, it is not to be placed back to the data set  $D$ .
- **Simple random sample with replacement (SRSWR) of size  $s$ :** This is similar to SRSWOR, except that each time a tuple is drawn from  $D$ , it is recorded and then replaced. That is, after a tuple is drawn, it is placed back in  $D$  so that it may be drawn again.





## 5. Data Transformation

### 5.4. Sampling

- **Cluster sample:** If the tuples in  $D$  are grouped into  $M$  mutually disjoint “clusters,” then a sample of  $s$  clusters can be obtained, where  $s < M$ .
- **Stratified sample:** If  $D$  is divided into mutually disjoint parts called *strata*, a stratified sample of  $D$  is generated by obtaining a sample at each stratum. This helps ensure a representative sample, especially when the data are skewed.
  - For example, a stratified sample may be obtained from customer data, where a stratum is created for each customer age group. In this way, the age group having the smallest number of customers will be sure to be represented.



## 6. Dimensionality Reduction

- Dimensionality reduction is the process of reducing the number of random variables or attributes or features under consideration.
- Dimensionality reduction methods
  - Principal components analysis
  - Attribute subset selection
  - Nonlinear methods: Kernel PCA, stochastic neighbor embedding



## 6. Dimensionality Reduction

### 6.1. Principal Component Analysis

- Suppose that the data to be reduced consist of tuples or data vectors described by  $d$  attributes or dimensions.
- **Principal components analysis** (PCA; also called the Karhunen-Loeve, or K-L, method) searches for  $k$   $d$ -dimensional orthonormal vectors that can best be used to represent the data, where  $k \leq d$ .
  - The original data are thus projected onto a much smaller space, resulting in dimensionality reduction.
  - PCA “combines” the essence of attributes by creating an alternative, smaller set of variables. The initial data can then be projected onto this smaller set.
  - PCA often reveals relationships that were not previously suspected and thereby allows interpretations that would not ordinarily result.



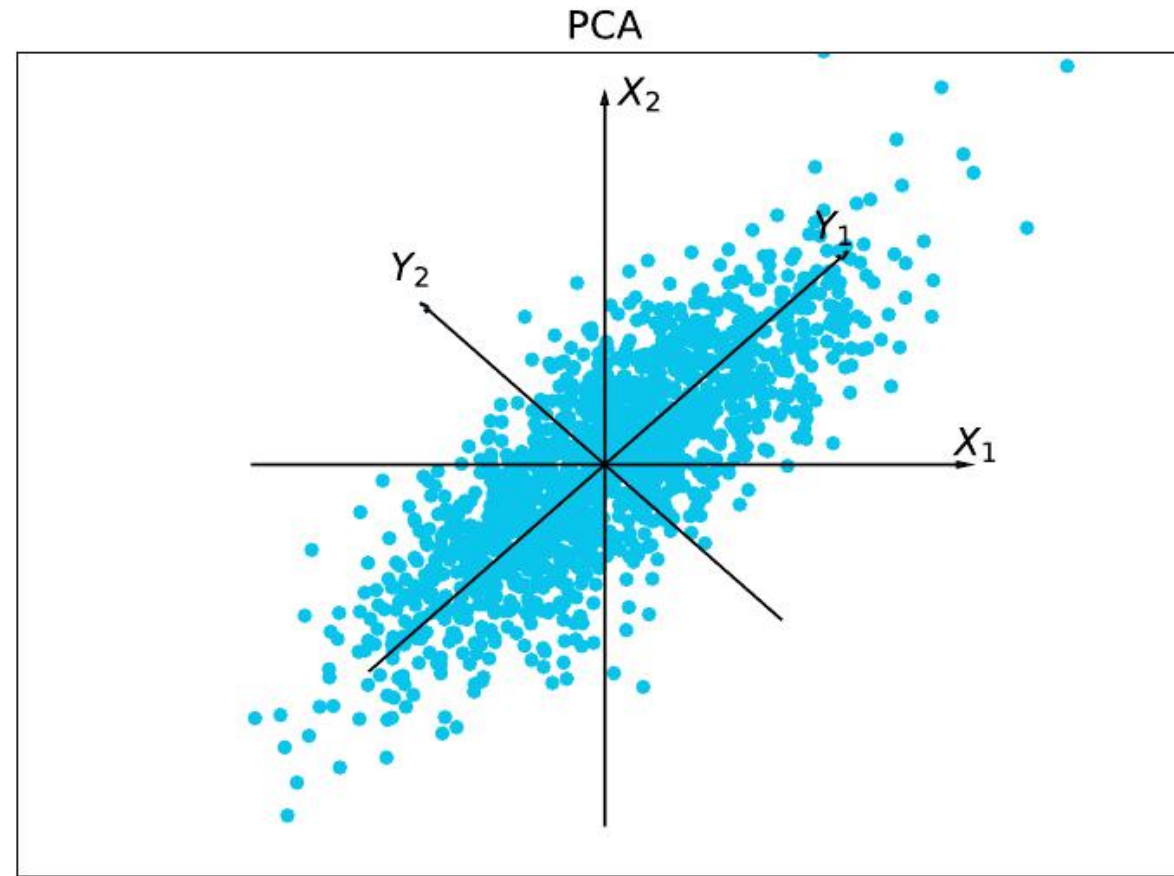
## 6. Dimensionality Reduction

### 6.1. Principal Component Analysis

- The basic procedure is as follows:
  1. The input data are normalized, so that each attribute falls within the same range. This step helps ensure that attributes with large domains will not dominate attributes with smaller domains.
  2. PCA computes  $k$  orthonormal vectors that provide a basis for the normalized input data. These are unit vectors that are perpendicular with each other. These vectors are referred to as the *principal components*. The input data are a linear combination of the principal components.
  3. The principal components are sorted in order of decreasing “significance” or strength. The principal components essentially serve as a new set of axes for the data, providing important information about variance.
    - That is, the sorted axes are such that the first axis shows the most variance among the data, the second axis shows the next highest variance, and so on. For example, Fig. 2.17 shows the first two principal components,  $Y_1$  and  $Y_2$ , for the given set of data originally mapped to the axes  $X_1$  and  $X_2$ . This information helps identify groups or patterns within the data.
  4. Because the components are sorted in descending order of “significance,” the data size can be reduced by eliminating the weaker components, that is, those with low variance. Using the strongest principal components, it should be possible to reconstruct a good approximation of the original data.

## 6. Dimensionality Reduction

### 6.1. Principal Component Analysis



**FIGURE 2.17**

Principal components analysis.  $Y_1$  and  $Y_2$  are the first two principal components for the given data.



## 6. Dimensionality Reduction

### 6.1. Principal Component Analysis

- PCA can be applied to ordered and unordered attributes and can handle sparse data and skewed data.
- Multidimensional data of more than two dimensions can be handled by reducing the problem to two dimensions.
- Principal components may be used as inputs to multiple regression and cluster analysis.



## 6. Dimensionality Reduction

### 6.2. Attribute Subset Selection

- Data sets for analysis may contain hundreds of attributes, many of which may be irrelevant to the mining task or redundant.
  - For example, if the task is to classify customers based on whether or not they are likely to purchase a popular new music album when notified of a sale, attributes such as the customer's phone number are likely to be irrelevant, unlike attributes such as *age* or *music\_taste*.
- **Attribute subset selection (feature subset selection)** reduces the data set size by removing irrelevant or redundant attributes (or dimensions). This makes mining focused on the relevant dimensions.
  - Mining on a reduced set of attributes has an additional benefit: It reduces the number of attributes appearing in the discovered patterns, helping to make the patterns easier to understand.
- *“How can we find a ‘good’ subset of the original attributes?”*



## 6. Dimensionality Reduction

### 6.2. Attribute Subset Selection

- “*How can we find a ‘good’ subset of the original attributes?*”
- For  $d$  attributes, there are  $2^d$  possible subsets. An exhaustive search for the optimal subset of attributes can be prohibitively expensive, especially as  $d$  and the number of data classes increase.
- Therefore, heuristic methods that explore a reduced search space are commonly used for attribute subset selection.
  - These methods are typically **greedy** in that, while searching through attribute space, they always make what looks to be the best choice at the time.
  - Their strategy is to make a locally optimal choice in the hope that this will lead to a globally good solution. Such greedy methods are effective in practice and may come close to estimating an optimal solution.





## 6. Dimensionality Reduction

### 6.3. Nonlinear Dimensionality Reduction Methods

- PCA is a linear method for dimensionality reduction in that each principal component is a linear combination of the original input attributes.
  - This works well if the input data approximately follows a Gaussian distribution or forms a few linearly separable clusters.
  - When the input data are linearly inseparable, however, PCA becomes ineffective. Luckily, there are many nonlinear methods we can resort to in this case.



## 6. Dimensionality Reduction

### 6.3. Nonlinear Dimensionality Reduction Methods

- **General Procedure**
- Suppose there are  $n$  data tuples  $\mathbf{x}_i$ , ( $i = 1, \dots, n$ ), each of which is represented by a  $d$ -dimensional attribute vector.
- How can we reduce the dimensionality to  $k$  where  $k \ll d$ ? In order words, we want to represent each of input data tuples by a  $k$ -dimensional attribute vector  $\hat{\mathbf{x}}_i$ , ( $i = 1, \dots, n$ ).
- Since  $k \ll d$ , we call the  $k$ -dimensional attribute vector  $\hat{\mathbf{x}}_i$ , ( $i = 1, \dots, n$ ) as low-dimensional representations of the original data tuples  $\mathbf{x}_i$ , ( $i = 1, \dots, n$ ).



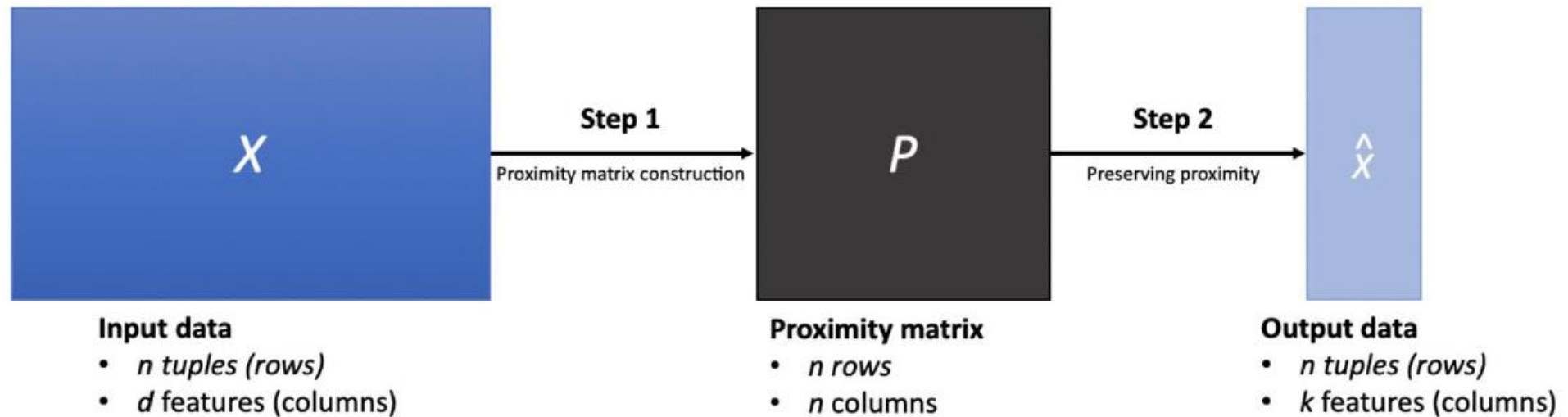
## 6. Dimensionality Reduction

### 6.3. Nonlinear Dimensionality Reduction Methods

- For many nonlinear dimensionality reduction methods, they often follow the following two steps.
- In the first step (*constructing proximity matrix*), we construct an  $n \times n$  proximity matrix  $P$  whose entry  $P(i, j)$  ( $i, j = 1, \dots, n$ ) indicates the affinity or relevance between the two corresponding data tuples  $\mathbf{x}_i$  and  $\mathbf{x}_j$ .
- In the second step (*preserving proximity*), we learn the new, low-dimensional representations of the input data tuples in the  $k$ -dimensional space  $\hat{\mathbf{x}}_i$  ( $i = 1, \dots, n$ ) so that the proximity matrix  $P$  constructed in the first step is somewhat preserved.

## 6. Dimensionality Reduction

### 6.3. Nonlinear Dimensionality Reduction Methods



**FIGURE 2.19**

An illustration of nonlinear dimensionality reduction.

## 6. Dimensionality Reduction

### 6.3. Nonlinear Dimensionality Reduction Methods

- Depending on how the proximity matrix is constructed (Step 1) and how to preserve the constructed proximity matrix (Step 2), a variety of nonlinear dimensionality reduction techniques have been developed.
  - Kernel PCA (KPCA)
  - Stochastic Neighbor Embedding (SNE).

**Table 2.8 Comparison of KPCA and SNE.**

|      | Step 1: Proximity Construction                                         | Step 2: Preserving Proximity                                                |
|------|------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| KPCA | $P(i, j) = \kappa(\mathbf{x}_i, \mathbf{x}_j)$                         | $\min \sum_{i,j=1}^n (P(i, j) - \hat{P}(i, j))^2 = \ P - \hat{P}\ _{fro}^2$ |
| SNE  | $P(i, j) = \frac{e^{-d_{ij}^2}}{\sum_{l=1, l \neq i}^n e^{-d_{il}^2}}$ | $\min \sum_{i=1}^n \text{KL}(P_i    \hat{P}_i)$                             |

## 6. Dimensionality Reduction

### 6.3. Nonlinear Dimensionality Reduction Methods

#### Kernel PCA

- In kernel PCA (KPCA), we use a *kernel function*  $\kappa(\cdot)$  to construct the proximity matrix called kernel matrix (Step 1):  $P(i, j) = \kappa(\mathbf{x}_i, \mathbf{x}_j)$ ,  $(i, j = 1, \dots, n)$ .
  - In the simplest term, a kernel function computes the similarity of a pair of input data tuples in some high-dimensional, often nonlinear, space.
- We can also estimate such proximity (i.e., similarity) based on the learned low-dimensional representations:  $\hat{P}(i, j) = \hat{\mathbf{x}}_i \cdot \hat{\mathbf{x}}_j$ ,  $(i, j = 1, \dots, n)$  where  $\cdot$  is the vector inner product.
- What would be the best (i.e., optimal) low-dimensional representations  $\hat{\mathbf{x}}_i$ ,  $(i = 1, \dots, n)$ ?
- Intuitively we hope that the estimated proximity matrix  $\hat{P}$  is as close as possible to the kernel matrix  $P$ . This leads to the following optimization problem (Step 2), which says that the best low-dimensional representations should be those that minimize  $\sum_{i,j=1}^n (P(i, j) - \hat{P}(i, j))^2 = \|P - \hat{P}\|_{fro}^2$ , where  $\|\cdot\|_{fro}$  is the matrix Frobenius norm.
- We will not go into the mathematical details of how to solve this optimization problem. To make the long story short, it turns out the optimal low-dimensional representations  $\hat{\mathbf{x}}_i$ ,  $(i = 1, \dots, n)$  can be obtained by the top- $k$  eigenvectors and eigenvalues of the kernel matrix  $P$ .

## 6. Dimensionality Reduction

### 6.3. Nonlinear Dimensionality Reduction Methods

#### Kernel PCA

- Typical choices for the kernel functions include
  - (1) Polynomial kernel:  $\kappa(\mathbf{x}_i, \mathbf{x}_j) = (1 + \mathbf{x}_i \cdot \mathbf{x}_j)^p$  where  $p$  is the parameter
  - (2) Radial basis function (RBF)  $\kappa(\mathbf{x}_i, \mathbf{x}_j) = e^{\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}}$ , where  $\sigma$  is the parameter.
- If we choose a linear kernel:  $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i \cdot \mathbf{x}_j$ , KPCA degenerates to the standard PCA.

# 6. Dimensionality Reduction

## 6.3. Nonlinear Dimensionality Reduction Methods

### Stochastic Neighbor Embedding

- We first construct the proximity matrix  $P$  as follows:

$$P(i, j) = \frac{e^{-d_{ij}^2}}{\sum_{l=1, l \neq i}^n e^{-d_{il}^2}}, \text{ where } d_{ij}^2 = \frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2} \text{ and } \sigma \text{ is the parameter.}$$

- We can view  $P(i, j)$  as the probability that data tuple  $\mathbf{x}_j$  is the neighbor of data tuple  $\mathbf{x}_i$ : the closer the two data tuples are (i.e., smaller  $d_{ij}$ ), the more likely  $\mathbf{x}_j$  is the neighbor of  $\mathbf{x}_i$ .
- Suppose we have learned the low-dimensional representations  $\hat{\mathbf{x}}_i$ , ( $i = 1, \dots, n$ ). We can obtain another estimated proximity matrix in the similar way:

$$\hat{P}(i, j) = \frac{e^{-\|\hat{\mathbf{x}}_i - \hat{\mathbf{x}}_j\|^2}}{\sum_{l=1, l \neq i}^n e^{-\|\hat{\mathbf{x}}_i - \hat{\mathbf{x}}_l\|^2}}$$

- In order to figure out the best low-dimensional representations  $\hat{\mathbf{x}}_i$ , ( $i = 1, \dots, n$ ), we again seek those that make the estimated proximity  $\hat{P}$  be as close as possible to the proximity matrix  $P$ :  $P \approx \hat{P}$ .
- $\hat{\mathbf{x}}_i = \arg \min_{\hat{\mathbf{x}}_i (i=1, \dots, n)} \sum_{j=1}^n D_{KL}(P_i || \hat{P}_j)$ , where  $P_i$  and  $\hat{P}_j$  are the  $i$ th rows of  $P$  and  $\hat{P}$ , respectively.